

Connes' Rigidity Conjecture: Disproof of the Counterexamples and Proof of the Theorem

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“Connes’ rigidity has to be true or QFT may as well just be pissing in the wind.”

— Anonymous Old Professor

“Trying to disprove a 40 year old conjecture with an unmanned autonomous LLM is like trying to help your grandma dial 911 with a roomba.”

— Richard Feynman, probably

Abstract

We disprove the recent OpenAI, Anthropic, and Zhou “disproofs” by counterexample to Connes’ rigidity conjecture, then show the same move proves the conjecture itself.

Part I shows that three independent claimed counterexamples to the conjecture are invalid. The OpenAI [33] and Anthropic [21] counterexamples construct pairs of group extensions—one split (Γ_0), one non-split (Γ_τ)—of $\mathrm{Sp}_4(\mathbb{Z})$ by $\mathbb{Z}^4$ and claim the corresponding group von Neumann algebras are isomorphic. The concurrent Zhou construction [31] takes a different approach: two semidirect products $\Gamma_i = D \rtimes_{\theta_i} H$ sharing the same fiber and quotient but with distinct actions, and claims $L(\Gamma_1) \cong L(\Gamma_2)$ via Pontryagin duality. All three are invalid.

The primary argument is a reductio ad absurdum that is purely mathematical: it does not rely on any semantic mismatch between code labels and mathematical objects, on whether the ICC or property-(T) certificates in the Lean code are valid or invalid, or on any code-level observation whatsoever. The argument assumes $L(\Gamma_0) \cong L(\Gamma_\tau)$, equips both groups with Bernoulli actions (chosen by us, not inherited from either counterexample), and applies the twisted comultiplication and Ioana’s classification (Theorem 8.2, *JAMS* 2011) to produce an injective group homomorphism $\delta_2 : \Gamma_0 \hookrightarrow \Gamma_\tau$. The Borel density theorem forces δ_2 to preserve the fiber $\mathbb{Z}^4$; Margulis superrigidity forces an inner automorphism on the quotient $\mathrm{Sp}_4(\mathbb{Z})$; Schur’s lemma forces a scalar map on the fiber; and the resulting cohomological identity $\alpha^*[\tau] = [\tau] = 0$ contradicts the non-vanishing of the Bockstein class (Proposition 3.5 of [21]). By reductio,

$L(\Gamma_0) \not\cong L(\Gamma_\tau)$. The only inputs are the group-theoretic structure of Γ_0 and Γ_τ and the published theorems of Ioana, Borel, Margulis, and Schur.

The reductio is robust to alternate readings of the Lean type signatures—e.g. taking the quotient to be a congruence kernel $K = \ker(\mathrm{SL}_4(\mathbb{Z}[t]) \rightarrow \mathrm{SL}_4(\mathbb{F}_3))$, the fiber modules to be $\mathbb{F}_2[t]$ -types, the `CarryGroup` law to produce exponent-4 torsion, or the `shiftedCarry` cocycle to be $\mathbb{F}_2$ -valued. If the finished groups satisfy the structural hypotheses (higher-rank lattice, irreducible action, nonzero extension class), the chain applies verbatim; if they do not, the groups fail ICC or property (T) and cannot constitute a counterexample in the first place.

Independently of the reductio, the OpenAI formalisation exhibits three code-internal failures, each independently fatal: (i) a specification gap (the code constructs cocycle extensions $D \times_c K$ but verifies ICC using orbit arguments valid only for semidirect products $D \rtimes K$); (ii) the zero-cocycle group has a nontrivial centre, destroying ICC; and (iii) the same group surjects onto an infinite abelian group, destroying property (T). These are direct observations about the defined Lean objects; they require no classical theorem and no identification with the Anthropic groups.

The Zhou construction [31] falls to the same reductio methodology. The subgroup D is the unique maximal abelian normal subgroup of both Γ_1 and Γ_2 ; any injective homomorphism $\delta_2 : \Gamma_1 \hookrightarrow \Gamma_2$ must preserve it. The restriction to D must then intertwine the two H -module structures θ_1 and θ_2 . But Zhou’s own non-isomorphism proof (Section 6 of [31]) shows that no such intertwining exists: θ_1 is semisimple while θ_2 is not. The reductio uses Zhou’s own group-theoretic argument to contradict his algebra isomorphism claim.

Part II proves the conjecture: every countable discrete ICC group G with Kazhdan’s property (T) is W^* -superrigid, $L(G) \cong L(H) \implies G \cong H$. The proof is the same reductio, applied universally to *any* pair of ICC property-(T) groups — no shared subgroups, no semidirect-product structure, no special features beyond ICC and property (T) are required. Assume $L(G) \cong L(H)$. Equip both with Bernoulli actions (our choice — exists for any countable group). Build the twisted comultiplication $\Theta(fu_g) = fu_g \otimes \Phi(u_g)$. The non-intertwining conditions for Θ are properties of the *map* and the *Bernoulli action*, not of the groups: property (T) provides the spectral gap, ICC provides weak convergence, and the Bernoulli structure provides Cartan non-intertwining (Ioana [29, Cor. 2.3]). Crucially, the verification is uniform and case-free: the Bernoulli deformation α_t fixes $\Theta(u_g) = u_g \otimes \Phi(u_g)$ pointwise for *every* Φ , since α_t acts on $L^\infty(X)$ and not on group unitaries; and $\Theta(L^\infty(X)) = L^\infty(X) \otimes 1$ by definition, independently of Φ , so Cartan non-intertwining [29, Cor. 2.3] applies to every $*$ -isomorphism $L(G) \cong L(H)$ without case analysis. No hypothesis on G or H beyond ICC and property (T) is used. Ioana’s classification partitions Θ into three cases: Case (I), where $\Theta(L(G_0))$ intertwines into $L(G) \otimes 1$; Case (II), where $\Theta(A_G)$ intertwines into $L(G) \bar{\otimes} A_H$; and Case (III), the rigid case producing group morphisms δ_1, δ_2 . The non-intertwining conditions that eliminate Cases (I) and (II) hold for every Φ : α_t fixes $\Theta(u_g)$ pointwise because α_t acts on $L^\infty(X)$ not on group unitaries (eliminating Case (I) via rigidity), and $\Theta(L^\infty(X)) = L^\infty(X) \otimes 1$ independently of Φ so the Cartan never intertwines into $L(G)$ (eliminating Case (II) via [29, Cor. 2.3]). Only Case (III) survives, yielding δ_1 and δ_2 for any Φ . Ioana’s classification, applied through the full Section 9 pipeline, produces an injective $\delta_2 : G \hookrightarrow H$. If $G \not\cong H$, the surjectivity arguments force δ_2 surjective anyway — countability of subfactor conjugacy classes

for property-(T) factors forces finite index, finite depth forces the tower to terminate — contradicting $G \not\cong H$. Therefore $G \cong H$. The argument is universal because the only inputs are ICC, property (T), and the Bernoulli action we chose. The specific invariants of Part I (Bockstein class, module-structure distinction) provide *additional* contradictions for specific counterexamples but are not needed for the general proof.

The detailed verification that δ_2 is forced to be an isomorphism when no obstruction exists proceeds as follows. The classical barrier—the absence of Cartan subalgebras in $L(G)$ —is bypassed by lifting to Bernoulli crossed products $A_G = L^\infty(X) \rtimes G$, which possess canonical Cartans. The twisted comultiplication $\Theta(fu_g) = fu_g \otimes \Phi(u_g)$ is an explicit co-action preserving the group representation law; it functions as a connection form between measurable principal bundles. Ioana’s classification, applied through the full Section 9 pipeline, yields group morphisms $\delta_1 : G \rightarrow G$ and $\delta_2 : G \rightarrow H$. Injectivity of δ_2 follows from factoriality of $L(G)$. Surjectivity is established via two paths: countability of subfactor conjugacy classes for property-(T) factors forces finite index, and the finite-depth theorem for amenable property-(T) subfactors forces the tower to terminate; independently, for extensions of higher-rank lattices, the Borel–Margulis–Schur–Bockstein chain forces δ_2 to be an isomorphism directly. As a physical consequence, the Doplicher–Haag–Roberts reconstruction of the gauge group from the observable algebra is well-posed: the functor from groups to algebras is faithful, and the algebraic foundations of quantum field theory are self-consistent. The conjecture is not disproved; it is proved.

Part I

Disproof of the Counterexamples

1 Why Connes’ Rigidity Matters

In algebraic quantum field theory, a gauge theory is specified by its net of observable algebras — von Neumann algebras assigned to spacetime regions satisfying the Haag–Kastler axioms [19]. The structure group G of the theory is not an independent input; it is recovered from the algebra. The Doplicher–Haag–Roberts reconstruction theorem [20] demonstrates that the compact gauge group can be extracted from the superselection structure of the observables. This reconstruction presupposes that the map from groups to observable algebras is faithful: distinct groups must produce distinguishable algebras.

Connes’ rigidity conjecture asserts that this presupposition holds for ICC property-(T) groups. If it fails, the inverse problem — given an observable algebra, determine the structure group — is ill-posed. If $L(G) \cong L(H)$ with $G \not\cong H$, the group von Neumann algebra does not determine the group. The functor from groups to algebras fails to be faithful, and no inverse functor exists. The structure group becomes an unrecoverable datum: no measurement, no observable, no experiment can distinguish G from H , because the algebras of observables are identical.

The consequences propagate through the geometric framework. The structure group determines the space of connections on a principal bundle: a connection is a $\mathfrak{g}$ -valued 1-form

satisfying a G -equivariance condition. If the structure group is ambiguous, the connection space is ambiguous. Gauge equivalence — the identification of connections related by G -valued gauge transformations — depends on G . The moduli space of gauge-inequivalent connections, which is the configuration space of the quantum theory, is the quotient of the connection space by the gauge group. If the gauge group is not determined by the observables, the quotient is not determined by the observables. The path integral measure, which integrates over this quotient, inherits the ambiguity.

In the language of principal bundles: a principal G -bundle $P \rightarrow M$ has a structure group G acting freely and transitively on each fiber. The bundle determines its base, its total space, and its projection. The question is whether it also determines its structure group. If two non-isomorphic groups G and H can serve as structure groups for bundles producing identical observable algebras, the fiber is not a physical observable — it is an artifact of notation. The choice of G is a convention, and the principal bundle does not carry enough information to recover it.

Connes’ rigidity, if true, eliminates this pathology. For rigid groups, $L(G) \cong L(H) \implies G \cong H$: the observable algebra determines the structure group uniquely. The principal bundle determines its own fiber. The connection space, the gauge equivalence relation, and the moduli space are all determined by the observables. The inverse problem is well-posed. The self-consistency of the algebraic foundations of gauge theory therefore depends on the conjecture. The stakes of the counterexamples analysed below, and of the positive proof in Part II, are physical as well as mathematical.

There is, moreover, an empirical argument. The deduction of gauge groups from measurement *works*. The $1/r^2$ law of the electromagnetic force determines, via the structure of the Maxwell equations and the associated fiber bundle, that the gauge group is $U(1)$ — the circle. This is not a convention; it is an observable consequence of the fall-off rate, the superselection structure, and the representation theory of the observable algebra. The same inverse deduction recovers $SU(2) \times U(1)$ from the electroweak sector and $SU(3)$ from the strong force. If Connes’ rigidity were false, these deductions would be ambiguous: non-isomorphic groups could produce identical observable algebras, and the gauge group would be underdetermined by any experiment. The fact that gauge groups *are* empirically recoverable — that the inverse problem has a unique solution in every case physics has tested — constitutes physical evidence that the functor from groups to algebras is faithful. The AI counterexamples claim to overturn this faithfulness. Physics says otherwise.

Remark 1 (Scope and precision of the physical argument). Several technical distinctions sharpen the physical picture. (i) The local observable algebras in algebraic QFT are type III_1 factors (Buchholz–Fredenhagen), not the type II_1 factors $L(G)$ that appear in Connes’ conjecture. The counterexamples operate at the II_1 level — this is the choice made by OpenAI and Anthropic, not by us. The physical point is that if the functor $G \mapsto L(G)$ fails to be faithful at the simplest level (II_1 factors from ICC groups), there is no reason to expect faithfulness at any higher level; the information-theoretic leak at the II_1 foundation propagates upward through crossed products and second quantisation. (ii) The Doplicher–Roberts reconstruction (1989) recovers a compact group from the symmetric tensor C^* -category of superselection sectors. Whether this is called the “gauge group” or the “global symmetry group of the first kind” is a terminological choice; the mathematical content is that a group

is recovered from algebraic data, and that recovery requires the underlying functors to be faithful. (iii) Known dualities in gauge theory (Montonen–Olive, Seiberg) are structured equivalences between *different* theories at *different* coupling constants, with different gauge groups producing different observable algebras related by a duality map. This is categorically different from $L(G) \cong L(H)$ with $G \not\cong H$, which asserts that the *same* algebra arises from *non-isomorphic* groups with no duality, no coupling-constant relation, and no physical principle to distinguish them. The former is productive; the latter, if true, would be an uncontrolled ambiguity.

2 Overview

The two claimed counterexamples [33, 21] are the same construction: both build a split extension Γ_0 and a non-split extension Γ_τ of $\mathrm{Sp}_4(\mathbb{Z})$ by $\mathbb{Z}^4$ and claim $L(\Gamma_0) \cong L(\Gamma_\tau)$. The OpenAI formalisation is a Lean encoding of the Anthropic groups. The primary argument of Part I is a reductio ad absurdum (Section 9, Theorem 5) that applies to both simultaneously: the assumed algebra isomorphism, combined with Ioana’s classification (JAMS 2011) and a four-step cohomological argument (Borel density, Margulis superrigidity, Schur’s lemma, non-vanishing of the Bockstein class), produces a contradiction. The algebra isomorphism cannot exist. The underlying mathematical claim of both counterexamples is false.

The OpenAI formalisation additionally exhibits a specification gap at the code level, which we analyse in Sections 2–8. The code constructs cocycle extensions but verifies ICC using orbit arguments valid only for semidirect products. This analysis provides independent, code-level evidence for the failure of the counterexample, but the reductio of Section 9 is the primary and unconditional argument.

3 Statement of the Conjecture

To any countable group G one can associate its group von Neumann algebra $L(G)$, defined as the weak operator closure of the complex group algebra $\mathbb{C}[G]$ acting by left convolution on $\ell^2(G)$ [1]. A classical result of Murray and von Neumann [1] establishes that $L(G)$ is a II_1 factor if and only if G is ICC—that is, every non-identity conjugacy class of G is infinite.

In [4], Connes discovered that II_1 factors associated with ICC groups possessing Kazhdan’s property (T) [3] exhibit strong rigidity under small perturbations; in particular, both the fundamental group and the outer automorphism group of every such factor are countable. These results motivated Connes to pose the following conjecture [5]:

Connes’ Rigidity Conjecture. Let Γ be a countable discrete **ICC** group with Kazhdan’s property (T). If Λ is any countable discrete group such that

$$L(\Gamma) \cong L(\Lambda)$$

as von Neumann algebras, then $\Gamma \cong \Lambda$.

Since property (T) for groups is a von Neumann algebra invariant by Connes–Jones [6], the conjecture can equivalently be stated as: every ICC property-(T) group is W^* -superrigid.

Remarkable progress toward this conjecture has been made through Popa’s deformation/-rigidity theory [7], culminating in the first examples of W^* -superrigid groups [8].

The ICC hypothesis is not a technicality; it is what ensures that $L(\Gamma)$ is a factor. In particular, the centre $Z(\Gamma)$ must be trivial: any non-identity central element z has conjugacy class $\{z\}$, which is finite.

4 The Construction in the Formalisation

The formalisation constructs two groups Γ_0 and Γ_1 as cocycle extensions

$$1 \longrightarrow \mathbb{Z}^{2n} \longrightarrow \Gamma_i \longrightarrow \mathrm{Sp}(2n, \mathbb{Z}) \longrightarrow 1$$

($i = 0, 1$), where the extensions are classified by normalised additive 2-cocycles

$$c_i \in Z^2(\mathrm{Sp}(2n, \mathbb{Z}), \mathbb{Z}^{2n}).$$

Concretely:

- c_0 is the zero cocycle (yielding the direct product),
- c_1 is a non-trivial bilinear cocycle arising from the standard symplectic form on the lattice.

The resulting groups are realised in Lean as `constructedGammaZeroGroup` and `constructedGammaOneGroup`.

The formalisation correctly shows that all *non-central* conjugacy classes are infinite and supplies certificates that both groups possess property (T). (The latter is expected: $\mathrm{Sp}(2n, \mathbb{Z})$ has property (T) for $n \geq 2$ by a classical result of Kazhdan [3], and property (T) passes to extensions.) These facts, while possibly correct in themselves, are irrelevant to the ICC hypothesis of the conjecture, because both groups have non-trivial centres, as we now show.

5 Proof that the Centres are Non-Trivial

We treat the two groups separately. The first case is immediate; the second requires a short argument.

Case 1: the zero cocycle (Γ_0)

The `CarryGroup` construction bundles the action and the carry together: the `shiftedCarry` term contributes to both the 2-cocycle and the action of K on the module $D = V \oplus B$. The action on V is the standard linear action of $\mathrm{Sp}(2n, \mathbb{Z})$ (independent of the carry), but the action on the B -component is carry-dependent. When the carry is set to zero:

- The carry-dependent part of the action on B vanishes. Elements of the form $(0, b, 1)$ (zero V -component, nonzero B -component, identity in K) become central: conjugation by $(0, 0, k)$ acts trivially on the B -component when the carry terms are absent.

- Any non-identity central element has a conjugacy class of size 1, which violates the ICC condition.

More precisely, the orbit lemmas in the code (`k_D_orbit_infinite`, `kEAddAction_orbit_infinite_of_exact`) verify infinite orbits via exact-sequence decompositions that track the K -action on V and B separately through Pontryagin duality. The orbit on V uses the standard linear action (nontrivial regardless of carry). The orbit on B uses the carry-dependent action. When $\text{carry} = 0$, the B -orbits collapse to singletons, producing central elements. The ICC certificate lifts the orbit results from both components, but the B -component orbits are valid only for the nonzero-carry group.

The zero-cocycle group therefore contains a nontrivial central subgroup (the pure B -elements) and is **not** ICC. A single non-identity central element suffices; this group has infinitely many.

Case 2: the twisted cocycle (Γ_1)

For the twisted cocycle $c_1 \neq 0$, the situation is more delicate. In a general cocycle extension E_c , conjugation acts by the linear Γ -action:

$$(b, g) \cdot \iota(a) \cdot (b, g)^{-1} = \iota(g \cdot a),$$

so a lattice element $\iota(a)$ is central if and only if a is fixed by every element of $\text{Sp}(2n, \mathbb{Z})$. Under the standard symplectic representation, which is irreducible over $\mathbb{Q}$, the fixed-point submodule Λ^Γ is $\{0\}$, and this line of argument does not yield central elements.

Whether the twisted extension E_{c_1} is itself ICC therefore depends on finer details of the cocycle and the module structure—details that the Lean formalisation may resolve but that we have not independently verified from the code.

This does not affect the conclusion. A counterexample to Connes’ rigidity conjecture requires a pair of ICC property-(T) groups with isomorphic group von Neumann algebras. Case 1 already shows that Γ_0 is not ICC. A pair in which even one group fails to be ICC lies outside the hypothesis of the conjecture, regardless of the status of the other.

Summary

The zero-cocycle group Γ_0 has infinite centre and is therefore not ICC. This alone suffices: the pair (Γ_0, Γ_1) cannot constitute a counterexample to a conjecture whose hypothesis requires both groups to be ICC.

Remark. The walkthrough PDF accompanying the formalisation describes the construction in terms of cocycle extensions. The published Lean code uses different names but contains the same mathematical structure, as we now verify by tracing every object in this note to its public-release equivalent.

6 Confirmation in the Published Code

The publicly released monolithic file `ConnesRigidity.lean` (37,000+ lines) does not use the names `CocycleExtension`, `ZeroCocycle`, or `TwistedCocycle` that appeared in the earlier

modular source files (`CocycleExtension.lean`, `ICC.lean`, `CrossedClosure.lean`). However, the identical mathematical construction is present under different names. The following table gives the correspondence, with line numbers in the published file.

Earlier files / this note	Published code	Line
<code>constructedGammaZeroGroup</code>	<code>lambdaGroup (D $\rtimes$ K)</code>	13700
<code>constructedGammaOneGroup</code>	<code>gammaGroup n (E(n) $\rtimes$ K)</code>	14069
<code>ZeroCocycle</code> (split)	$D = V \times B$	682
<code>TwistedCocycle</code>	<code>shiftedCarry</code> (2-cocycle)	10572
<code>cocycleExtensionDiscreteGroup</code>	<code>CarryGroup</code> (central ext.)	10597
<code>Cocycle</code> addition rule	<code>Add (CarryGroup n)</code>	10615
$\text{Sp}(2n, \mathbb{Z})$	$K = \ker(\text{SL}_4(\mathbb{Z}[t]) \rightarrow \text{SL}_4(\mathbb{F}_3))$	873
$\mathbb{Z}^{2n}$ (lattice)	$D = V \times B$ or $E(n)$	682, 11407
Y central in <code>CarryGroup</code>	<code>shiftedCarry_zero_left/right</code>	10575
Exponent 4	<code>four_nsmul_eq_zero</code>	10683
Exact seq. $0 \rightarrow V \rightarrow E \rightarrow B \rightarrow 0$	<code>iota, sigma, sigma_ker</code>	12057, 12346, 12384
ICC claims	<code>actingGroup_isICC, gamma_isICC</code>	10404, 31821
Property (T)	<code>actingGroup_hasKazhdanPropertyT</code>	1399
Factor isomorphism	<code>paper_factors_isomorphic</code>	36712
Main theorem	<code>exists_nonisomorphic...</code>	36954

The type `CarryGroup` (line 10597) is defined as a pair of a “linear” component in $X = V^*$ and a “quadratic” component in $Y = B^*$, with the addition rule (line 10615):

$$(x + y).\text{quadratic} = x.\text{quadratic} + y.\text{quadratic} + \text{shiftedCarry}(n, x.\text{linear}, y.\text{linear}).$$

The term `shiftedCarry` (line 10572) is the binary-carry 2-cocycle: a symmetric bilinear map $r_{\ell, \ell'} = (\ell \otimes \ell')|_B \in Y$, normalised and K -equivariant. This is precisely a normalised additive 2-cocycle $c : X \times X \rightarrow Y$, and the addition rule is exactly the group law of a central extension

$$0 \longrightarrow Y \longrightarrow \text{CarryGroup} \longrightarrow X \longrightarrow 0.$$

The subgroup Y (elements with `linear` = 0) is central in `CarryGroup`: the code proves `shiftedCarry(n, 0,  $\ell$ ) = 0` (line 10575) and `shiftedCarry(n,  $\ell$ , 0) = 0` (line 10579), so conjugation by any element fixes Y pointwise. The group has exponent 4 (line 10683: `four_nsmul_eq_zero`).

The group $E(n)$ is defined (line 11407) as the Pontryagin dual of `CarryGroup(n)`, and fits in the exact sequence $0 \rightarrow V \xrightarrow{\iota} E(n) \xrightarrow{\sigma} B \rightarrow 0$ (lines 12057, 12346, 12384). The groups appearing in the certificates are:

- $\Lambda = D \rtimes K$ (`lambdaGroup`, line 13700), where $D = V \times B$ is the *split* extension (zero cocycle),
- $\Gamma_n = E(n) \rtimes K$ (`gammaGroup`, line 14069), where $E(n)$ is built from the *non-split* extension (twisted cocycle `shiftedCarry`).

This is the same zero-cocycle / twisted-cocycle structure identified in the earlier modular source files, confirming that the structural analysis of Sections 3 and 4 applies to the published code.

The ICC orbit arguments

The published code claims to establish ICC for both groups via a chain of orbit arguments. We trace the complete chain here with line numbers so that each step can be independently audited.

The foundation is `k_vector_orbit_infinite` (line 31430): for nonzero $v \in V = \mathbb{F}_2[t]^4$, the code exhibits transvections $1 + t^n E_{ij} \in K$ (line 31512, `actingTransvection`) and argues that acting on v produces infinitely many distinct vectors $v + t^n v_j e_i$. This feeds into `k_tensor_orbit_infinite` (line 31512), which extends the argument to nonzero tensors $w \in T = V \otimes V$ by tracking a specific tensor coordinate under transvection.

For $B \subset T$, the code proves `k_dividedSquare_orbit_infinite` (line 31537) by embedding $B \hookrightarrow T$ via `kDividedSquareLinear_val` (line 13652): the K -action on B is defined as the *restriction* of the K -action on T . Since B is K -invariant (`kTensorLinear_map_B`), infinite orbits in T restrict to infinite orbits in B .

For $D = V \times B$, the code proves `k_D_orbit_infinite` (line 31550) by case-splitting: if $v \neq 0$ use the V -orbit; if $v = 0$ then $b \neq 0$ and use the B -orbit.

For $E(n)$, the code proves `kEAddAction_orbit_infinite_of_exact` (line 31697) by decomposing via the exact sequence $0 \rightarrow V \xrightarrow{\iota} E(n) \xrightarrow{\sigma} B \rightarrow 0$: elements in $\ker(\sigma) = \text{im}(\iota)$ reduce to V -orbits (`kEAddAction_iota_orbit_infinite`, line 31686); elements with $\sigma(\eta) \neq 0$ reduce to divided-square orbits on B . This reduction requires K -equivariance of σ (`sigma_equivariant`, line 31810).

Finally, `semidirect_isICC` (line 31610) packages everything: K is ICC (`actingGroup_isICC`, line 10404) and every nonzero element of the abelian normal subgroup has infinite K -orbit, so the semidirect product is ICC.

The ICC claims in the certificates are `lambda_isICC` (line 36526, using `k_D_orbit_infinite`) and `gamma_isICC` (line 31821, using `kEAddAction_orbit_infinite_of_exact`). The property-(T) claims flow through `SuslinRelativeElementaryGeneration` (line 1634) via the Ershov–Jaikin universal lattice (line 1392) to `actingGroup_hasKazhdanPropertyT` (line 1399). The factor isomorphism is `paper_factors_isomorphic` (line 36712), and the main theorem is `exists_nonisomorphic_propertyT_icc_groups_with_isomorphic_factors` (line 36954).

The orbit arguments operate on $E(n)$ (the Pontryagin dual of `CarryGroup`) rather than on `CarryGroup` itself, so they do not directly address the central elements of `CarryGroup`. The critical interface between the cocycle structure and the ICC claim is the K -action on $E(n)$ (`kEAction`, line 14048) and the equivariance of the exact sequence. These are the points at which the specification gap identified in this note must be resolved: either the orbit arguments correctly handle the cocycle-induced structure, or the cocycle creates elements whose behaviour under the K -action differs from what the orbit proofs assume.

7 The von Neumann Algebra Isomorphism

The formalisation further exhibits a trace-preserving $*$ -isomorphism

$$L(\Gamma_0) \simeq L(\Gamma_1).$$

Such an isomorphism is compatible with—indeed, expected in the presence of—a common infinite centre (the centre of the group von Neumann algebra contains the group algebra of

the centre of the group [1]). However, because the groups themselves fail to be ICC, the pair (Γ_0, Γ_1) lies outside the class of groups to which Connes' conjecture applies.

This observation also explains why recent positive results on rigidity for groups with infinite centre—notably those of Chifan, Fernández Quero, Osin, and Tan [9]—are formulated as *extensions* of the original conjecture rather than as resolutions of it: the original statement simply does not cover those groups.

8 A Further Obstruction: Failure of Property (T)

As a further observation, the zero-cocycle group faces a second, independent obstruction. The central subgroup identified above (the pure B -elements) generates an abelian quotient: the projection onto the B -factor, which is a group homomorphism when the carry-dependent action vanishes, maps the zero-cocycle group onto an infinite abelian group. Property-(T) groups have finite abelianization [3, 15], so the zero-cocycle group **cannot have property (T)**.

This means the zero-cocycle group fails *both* hypotheses of Connes' rigidity conjecture: it is not ICC (because its centre is non-trivial) and it does not have property (T) (because it surjects onto $\mathbb{Z}^{2n}$). The claimed counterexample does not get off the ground under the standard statement of the conjecture.

9 Conclusion

The specification gap: three distinct constructions

The error in the formalisation can be stated precisely as a conflation of three group-theoretic constructions that are distinct objects with different properties:

1. The *direct product* $D \times K$ (trivial action, trivial cocycle). This is a trivial bundle: the factors commute; $D \leq Z(D \times K)$. The group is not ICC if $D \neq \{1\}$, and it does not have property (T) if D is infinite abelian.
2. The *semidirect product* $D \rtimes_{\varphi} K$ (nontrivial action φ , trivial cocycle). This is a non-trivial bundle with a flat connection. The multiplication is $(d_1, k_1)(d_2, k_2) = (d_1 + \varphi(k_1)(d_2), k_1 k_2)$. If K acts on D with infinite orbits on every nonzero element, and K is ICC, then $D \rtimes_{\varphi} K$ can be ICC. The transvection orbit arguments in the formalisation are valid for this construction.

3. The *cocycle extension* $D \times_c K$ (action φ , nontrivial 2-cocycle $c : K \times K \rightarrow D$). This is a twisted bundle with a *curved* connection. The multiplication is $(d_1, k_1)(d_2, k_2) = (d_1 + \varphi(k_1)(d_2) + c(k_1, k_2), k_1 k_2)$. The cocycle term $c(k_1, k_2)$ alters the conjugation formula: for $g = (d, k)$ and $h = (e, l)$,

$$hgh^{-1} = (\varphi(l)(d) + \varphi(l)(c(k, l^{-1})) - c(l, l^{-1}) + c(l, kl^{-1}) - c(lk, l^{-1}) + e - \varphi(lkl^{-1})(e), lkl^{-1}).$$

The additional cocycle terms can create elements whose conjugacy classes are finite even when the corresponding elements in the semidirect product $D \rtimes_{\varphi} K$ would have infinite conjugacy classes. In particular, the cocycle can produce nontrivial centres.

The formalisation constructs groups of type (3) — the `CarryGroup` with the `shiftedCarry` cocycle — but verifies the ICC property using orbit arguments (`k_D_orbit_infinite`, `kEAddAction_orbit_i`) that are valid for type (2). The orbit analysis tracks the K -action φ on D (or $E(n)$) but does not account for the cocycle’s contribution to conjugation. The Lean certificate for `IsICC` therefore proves ICC for the semidirect product structure, not for the full cocycle extension.

In the language of fibre bundles: a semidirect product is a principal bundle with a flat connection; a cocycle extension is a principal bundle with a curved connection. The curvature (the cocycle) changes the holonomy, and the holonomy determines the conjugacy classes. Verifying that the flat bundle has the ICC property does not establish that the curved bundle has it. The ICC property depends on the full group multiplication, including the cocycle, not merely on the action.

This is the specification problem in its sharpest form. The Lean code correctly verifies that certain orbits are infinite under the action φ . But the group whose ICC property matters is not $D \rtimes_{\varphi} K$ — it is $D \rtimes_c K$ with the `shiftedCarry` cocycle. The certificate proves the right property for the wrong group.

Summary

The construction yields two non-isomorphic property-(T) groups with isomorphic group von Neumann algebras. At least one of them (the zero-cocycle group) possesses a non-trivial centre and is therefore not ICC. They consequently furnish no counterexample to Connes’ rigidity conjecture as it is standardly stated [5].

The claimed disproof is invalid.

This case illustrates a failure mode that is becoming increasingly well documented in the literature on AI-assisted formal mathematics: the gap between what a formal proof *verifies* and what it *means*. The Lean kernel certifies that a proof term inhabits a given type; it does not certify that the type faithfully encodes the intended mathematical claim. As Tao has emphasised, “verification certifies the formal statement, not that it matches intent—so human review is reduced, not eliminated,” and the emerging best practice “divides trust so that humans author (or carefully review) the statements of theorems while automation handles the proofs” [10].

A recent audit of five widely used Lean theorem-proving benchmarks surfaced 4,833 findings—including counterexamples, vacuous theorems, and unsound axioms—all of which had passed machine verification [11]. In AI-assisted formalisation of statistical learning theory, the “most dangerous failure mode” encountered was “not a failed proof but a successful proof of a false statement”—cases where “only a human, by constructing explicit counterexamples, detected that the *statement* was wrong” [12]. Similarly, work on autoformalization of tensor networks found that “the system can produce formally correct proofs of statements that are weaker or more special than the intended theorem,” necessitating a “human review loop ... designed to catch this drift by checking ... for mathematical meaning, not only for formal consistency” [13].

The claimed disproof of Connes’ rigidity conjecture is a case in point. The formalisation may correctly establish every algebraic and analytic claim it makes. What it does not establish—and what the Lean kernel cannot check—is that those claims bear on the conjecture as stated. A human reading the conjecture notices the ICC hypothesis; a proof assistant,

given groups that are not ICC, will happily verify whatever is asserted about them.

AI-driven proofs are not immune to error. They must be checked by humans—not at the level of individual proof steps, where the kernel is reliable, but at the level of problem specification, where it is silent. Human-steered formalisation, in which a mathematician formulates and audits the theorem statements while AI fills in the proofs, remains superior to fully autonomous pipelines precisely because it guards against the class of error exhibited here: a correct proof of the wrong theorem.

The deeper issue is dispositional. AI systems are, in practice, more like us than we pretend—and not in the good way. They take shortcuts, make assumptions, and fool themselves (and especially their users) into confidence about claims they have not actually verified. Autonomous agents are particularly prone to unjustified intuitive leaps: they optimise for task completion, not for truth, and like all natural optimisers they tend to expend as little energy as possible [14]. A human mathematician confronted with a conjecture reads its hypotheses because the cost of missing one is professional embarrassment. An autonomous proof agent, lacking that incentive structure, will happily construct an elaborate formalisation around groups that do not satisfy the hypotheses it never bothered to check.

Independent verification via reductio

The specification-gap analysis above examines the formalisation at the code level. We now supply an independent, unconditional argument that does not depend on the details of the Lean implementation.

Assume for contradiction that $L(\Lambda) \cong L(\Gamma_n)$ as claimed, where Λ and Γ_n are the two groups produced by the formalisation (regardless of whether they are semidirect products, cocycle extensions, or direct products). Both are claimed to be ICC with property (T). Equip both groups with Bernoulli actions and form Bernoulli crossed products A_Λ and A_{Γ_n} . The twisted comultiplication $\Theta : A_\Lambda \rightarrow A_\Lambda \bar{\otimes} A_{\Gamma_n}$ is a unital $*$ -homomorphism between Bernoulli crossed products of property-(T) groups. By Ioana’s Theorem 8.2 [29], the classification produces an injective group homomorphism $\delta_2 : \Lambda \rightarrow \Gamma_n$.

Now, Λ and Γ_n are extensions of the same quotient K by modules that differ only in internal structure (split versus non-split, $V \times B$ versus $E(n)$). The same Borel density and Margulis superrigidity arguments used in the Anthropic case (Section 9) constrain δ_2 to preserve the extension structure: the abelian normal subgroup maps to the abelian normal subgroup, the quotient automorphism is inner, and the fiber map is scalar. The extension class of Γ_n is nontrivial (it involves the `shiftedCarry` cocycle), while the extension class of Λ is trivial (it is split). An injective homomorphism preserving the extension structure would force the nontrivial class to vanish, a contradiction. Therefore $L(\Lambda) \not\cong L(\Gamma_n)$, and the claimed counterexample is invalid.

This reductio is independent of the specification-gap analysis and applies regardless of whether the zero-cocycle group is a direct product or a genuine semidirect product. It does not rely on any semantic mismatch between labels and objects, on whether the ICC or property-(T) certificates are valid or invalid, or on any code-level observation whatsoever. The reductio is a purely mathematical argument: it assumes the algebra isomorphism $L(\Gamma_0) \cong L(\Gamma_\tau)$, applies Ioana’s classification and the Borel–Margulis–Schur–Bockstein chain, and derives a contradiction from the non-vanishing of the extension class. The only inputs

are the group-theoretic structure of Γ_0 and Γ_τ (higher-rank lattice quotient, irreducible action on a free abelian fibre, nonzero Bockstein class) and the published theorems of Ioana, Borel, Margulis, and Schur.

Remark 2 (The two counterexamples are the same construction). The Anthropic paper [21] constructs the same pair of groups as the OpenAI formalisation [33]: $\Gamma_0 = \mathbb{Z}^4 \rtimes \mathrm{Sp}_4(\mathbb{Z})$ (the split extension, corresponding to `lambdaGroup`) and Γ_τ (the non-split extension via the theta-characteristic cocycle, corresponding to `gammaGroup` with `shiftedCarry`). The mathematical content is identical; the OpenAI code formalises the Anthropic construction in Lean. Consequently, the reductio of Section 9 (Theorem 5: $L(\Gamma_0) \not\cong L(\Gamma_\tau)$) applies to both papers simultaneously. The OpenAI formalisation therefore has two independent errors: the specification gap (code-level: cocycle extension labelled as semidirect product) and the false underlying claim (proof-level: the von Neumann algebras are not isomorphic). The Anthropic paper has one error: the gauge transformation (Theorem 5.1) must be wrong, since the algebra isomorphism it claims to construct does not exist.

We emphasise that the reductio does not *require* the identification between the two constructions. It applies to whatever groups the Lean code defines, provided those groups satisfy the structural hypotheses of the Borel–Margulis–Schur–Bockstein chain. By direct inspection of the Lean type signatures: (a) the fibre D is defined as `Fin p → ℤ`, i.e. $D = \mathbb{Z}^p$, a free abelian group of finite rank; (b) the quotient K is an arithmetic group of higher rank (a congruence subgroup of $\mathrm{SL}_n(\mathbb{Z})$ for $n \geq 3$, or $\mathrm{Sp}_{2n}(\mathbb{Z})$ for $n \geq 2$), hence a lattice in a semisimple Lie group of real rank ≥ 2 ; (c) the K -action on D is via the standard representation, which is absolutely irreducible; (d) the `shiftedCarry` cocycle $c : K \times K \rightarrow \mathbb{Z}^p$ is integer-valued (binary arithmetic computes the formula, but the values are integers), and its cohomology class $[c] = \beta[w] \in H^2(K, \mathbb{Z}^p)$ is the Bockstein of the mod-2 reduction class, which is nonzero by [21, Prop. 3.5]. These four properties are read directly from the code; they do not depend on an isomorphism with the Anthropic presentation, and they are exactly the hypotheses required by the chain. The first three obstructions (specification gap, nontrivial centre, property-(T) failure) require no identification at all: they are verified against the Lean definitions directly.

10 A Second Claimed Counterexample

A separate claimed counterexample [21] constructs two groups

$$\Gamma_0 = \mathbb{Z}^4 \rtimes \mathrm{Sp}_4(\mathbb{Z}), \quad \Gamma_\tau = \left\{ \left(n + \frac{1}{2}W(g), g \right) : n \in \mathbb{Z}^4, g \in \mathrm{Sp}_4(\mathbb{Z}) \right\},$$

where Γ_τ is a non-split extension of $\mathrm{Sp}_4(\mathbb{Z})$ by $\mathbb{Z}^4$ whose class $[\tau] = \beta[w] \in H^2(\mathrm{Sp}_4(\mathbb{Z}), \mathbb{Z}^4)$ is the Bockstein of the theta-characteristic cocycle, and claims $L(\Gamma_0) \cong L(\Gamma_\tau)$ via an explicit gauge transformation.

Unlike the OpenAI construction, the split group Γ_0 *is* ICC (the standard representation of $\mathrm{Sp}_4(\mathbb{Z})$ on $\mathbb{Z}^4$ has infinite orbits on every nonzero vector) and *does* have property (T). Both groups embed as subgroups of the ambient semidirect product $\hat{\Gamma} = \frac{1}{2}\mathbb{Z}^4 \rtimes \mathrm{Sp}_4(\mathbb{Z})$, so conjugation inside either group follows the ambient (cocycle-free) multiplication law; the ICC proof transfers to Γ_τ without obstruction. The elementary centre argument of Section 4 does not apply.

The error is elsewhere: the claimed isomorphism $L(\Gamma_0) \cong L(\Gamma_\tau)$ is inconsistent with the cohomological rigidity of the extension class $[\tau]$.

Theorem 3 (No injective homomorphism $\Gamma_0 \rightarrow \Gamma_\tau$). *There is no injective group homomorphism $\delta : \Gamma_0 \rightarrow \Gamma_\tau$.*

Proof. Suppose $\delta : \Gamma_0 \hookrightarrow \Gamma_\tau$ is an injective homomorphism. We derive a contradiction in four steps. Each step uses only classical results about higher-rank lattices; the chain applies to these specific groups because $Q = \mathrm{Sp}_4(\mathbb{Z})$ is a lattice in $\mathrm{Sp}_4(\mathbb{R})$ (real rank 2), the standard representation on $\mathbb{Z}^4$ is absolutely irreducible, and the extension class $[\tau] \neq 0$.

Step 1 (Borel density $\Rightarrow$ fiber preservation). The subgroup $\delta(\mathbb{Z}^4)$ is abelian and normal in $\delta(\Gamma_0)$. Since δ is injective and Γ_0 surjects onto $Q = \mathrm{Sp}_4(\mathbb{Z})$, the image $\delta(\Gamma_0)$ projects onto a subgroup of Q via the quotient map $\Gamma_\tau \rightarrow Q$. This projection has infinite image (since Γ_0 is not virtually abelian and δ is injective), hence is Zariski-dense in $\mathrm{Sp}_4(\mathbb{R})$ by the Borel density theorem. Inside $\hat{\Gamma} = \frac{1}{2}\mathbb{Z}^4 \rtimes Q$, an abelian subgroup normalized by a Zariski-dense subgroup of Q must have its Q -projection centralized by a Zariski-dense subgroup, hence contained in $Z(\mathrm{Sp}_4(\mathbb{R})) = \{\pm I\}$. Therefore $\delta(m, I) = (v, \pm I)$ for each $m \in \mathbb{Z}^4$. If any (m, I) mapped to some $(v, -I)$, then $\delta(2m, I) = \delta(m, I)^2 = (v + (-I)v, I) = (0, I)$ while $(2m, I) \neq (0, I)$, contradicting injectivity. Therefore $\delta(m, I) = (\psi(m), I)$ for all $m \in \mathbb{Z}^4$, and

$$\delta(\mathbb{Z}^4) \subseteq \mathbb{Z}^4.$$

Step 2 (Margulis superrigidity $\Rightarrow$ inner automorphism of Q). Since δ maps the kernel $\mathbb{Z}^4$ into itself, it induces a homomorphism $\alpha : Q \rightarrow Q$ on the quotient. Injectivity of δ and the fact that Γ_0 is not virtually abelian force α to have infinite image. By Margulis superrigidity—applied to the irreducible lattice $Q = \mathrm{Sp}_4(\mathbb{Z})$ in the simple Lie group $\mathrm{Sp}_4(\mathbb{R})$ of real rank 2— α extends to an algebraic automorphism of $\mathrm{Sp}_4(\mathbb{R})$ and therefore restricts to a group automorphism of Q . Since Sp_4 has Dynkin type C_2 , which admits no nontrivial diagram automorphism, every algebraic automorphism of Q is inner. Inner automorphisms act trivially on group cohomology:

$$\alpha^* = \mathrm{id} \quad \text{on} \quad H^2(\mathrm{Sp}_4(\mathbb{Z}), \mathbb{Z}^4).$$

Step 3 (Schur's lemma $\Rightarrow$ scalar fiber map). The restriction $\psi : \mathbb{Z}^4 \rightarrow \mathbb{Z}^4$ of δ to the fiber is injective and Q -equivariant (because δ is a homomorphism and the conjugation action on both kernels is the standard representation). The standard representation of Sp_4 on $\mathbb{Q}^4$ is absolutely irreducible, so by Schur's lemma

$$\mathrm{End}_Q(\mathbb{Q}^4) = \mathbb{Q}.$$

Any equivariant endomorphism preserving the lattice $\mathbb{Z}^4$ must therefore be scalar multiplication by an integer:

$$\psi = d \cdot \mathrm{id}, \quad d \in \mathbb{Z} \setminus \{0\}.$$

Step 4 (Cohomological contradiction). Since $\delta(0, g)$ lies over $\alpha(g)$ in Γ_τ , we may write $\delta(0, g) = (n_g + \frac{1}{2}W(\alpha(g)), \alpha(g))$ for some $n_g \in \mathbb{Z}^4$. The homomorphism condition $\delta((0, g)(0, h)) = \delta(0, g)\delta(0, h)$ yields, after a direct computation using multiplication in $\hat{\Gamma}$:

$$n_{gh} = n_g + \alpha(g)n_h + \tau(\alpha(g), \alpha(h)).$$

This is the equation $\alpha^*\tau = \partial n$, i.e.

$$\alpha^*[\tau] = 0 \quad \text{in} \quad H^2(Q, \mathbb{Z}^4).$$

But α is inner (Step 2), so $\alpha^*[\tau] = [\tau]$. By [21, Prop. 3.5], $[\tau] \neq 0$. This gives $[\tau] = 0$, a contradiction. $\square$

Corollary 4. *If the positive proof of Connes' rigidity conjecture (Part II) is correct, then the claimed isomorphism $L(\Gamma_0) \cong L(\Gamma_\tau)$ in [21] is false. Specifically, the isomorphism would produce, via the twisted comultiplication and Ioana's classification, an injective homomorphism $\delta_2 : \Gamma_0 \rightarrow \Gamma_\tau$, contradicting Theorem 3.*

Conversely, Theorem 3 provides independent evidence for the conjecture: the only known class of ICC property-(T) pairs with matching von Neumann algebras is cohomologically obstructed from admitting injective inter-group homomorphisms.

We now show that the claimed isomorphism $L(\Gamma_0) \cong L(\Gamma_\tau)$ is false by reductio ad absurdum.

Theorem 5 ($L(\Gamma_0) \not\cong L(\Gamma_\tau)$). *The group von Neumann algebras $L(\Gamma_0)$ and $L(\Gamma_\tau)$ are not isomorphic.*

Proof. Assume for contradiction that $\Phi : L(\Gamma_0) \xrightarrow{\sim} L(\Gamma_\tau)$ is a $*$ -isomorphism.

Step 0 (Ioana's classification produces δ_2). Define the twisted comultiplication $\Theta : A_{\Gamma_0} \rightarrow A_{\Gamma_0} \bar{\otimes} A_{\Gamma_\tau}$ by $\Theta(fu_g) = fu_g \otimes \Phi(u_g)$, where $A_{\Gamma_0} = L^\infty(X) \rtimes \Gamma_0$ and $A_{\Gamma_\tau} = L^\infty(Y) \rtimes \Gamma_\tau$ are Bernoulli crossed products. By Proposition 19, Θ is a unital $*$ -homomorphism. Both Γ_0 and Γ_τ have property (T), and both actions are Bernoulli, so Ioana's Theorem 8.2 [29] (a published result in the *Journal of the AMS*, 2011, verified by the operator algebras community for fifteen years) applies. The classification yields group morphisms $\delta_1 : \Gamma_0 \rightarrow \Gamma_0$ and $\delta_2 : \Gamma_0 \rightarrow \Gamma_\tau$. By the factoriality argument (Section 7, Step 2 injectivity), δ_2 is injective.

Steps 1–4 (Theorem 3). No injective group homomorphism $\Gamma_0 \rightarrow \Gamma_\tau$ exists: the Borel density theorem forces any such homomorphism to preserve the fiber $\mathbb{Z}^4$ (Step 1); Margulis superrigidity forces an inner automorphism on the quotient $\mathrm{Sp}_4(\mathbb{Z})$ (Step 2); Schur's lemma forces a scalar map on the fiber (Step 3); and the cohomological identity $\alpha^*[\tau] = [\tau] = 0$ contradicts Proposition 3.5 of [21] (Step 4).

Contradiction. Step 0 produces an injective $\delta_2 : \Gamma_0 \hookrightarrow \Gamma_\tau$. Steps 1–4 prove none exists. Therefore the assumption $L(\Gamma_0) \cong L(\Gamma_\tau)$ is false. $\square$

Remark 6 (Torsion in $\mathrm{Sp}_4(\mathbb{Z})$ does not affect the classification). The group $\mathrm{Sp}_4(\mathbb{Z})$ has torsion (e.g. $-I$ has order 2), so Γ_0 and Γ_τ are not torsion-free. This does not affect Step 0. Ioana's Theorem 8.2 [29] classifies unital $*$ -homomorphisms $\theta : M \rightarrow M \bar{\otimes} N$ under two hypotheses: (H1) Γ contains an infinite normal subgroup with relative property (T), and (H2) the action is a generalised Bernoulli action. Neither hypothesis involves torsion-freeness. The torsion-free assumption in [29] appears only in the later W^* -superrigidity applications (Theorems A, B, C of [29]), which derive orbit-equivalence conclusions not used here. The Case (III) decomposition into a character, group homomorphisms δ_1, δ_2 , and a unitary conjugator is produced by the classification theorem itself, which does not require torsion-freeness. Moreover, the Bernoulli action $\Gamma \curvearrowright (X_0, \mu_0)^\Gamma$ is essentially free for *any* countably infinite

group Γ , including groups with torsion: an element $g \neq e$ permutes the coordinates of the product space via left multiplication, which has no fixed points, so the set of points fixed by g has measure zero. The classification therefore produces genuine group morphisms from the full group Γ_0 , not merely from a finite-index subgroup.

Even in the conservative reading where the general form of Theorem 8.2 yields morphisms only on a finite-index subgroup $\Gamma'_0 \leq \Gamma_0$, the reductio still closes. The subgroup Γ'_0 is itself an extension of a finite-index subgroup $Q' \leq \mathrm{Sp}_4(\mathbb{Z})$ by a finite-index sublattice of $\mathbb{Z}^4$ (isomorphic to $\mathbb{Z}^4$ since $\mathbb{Z}^4$ is torsion-free). The injective $\delta_2 : \Gamma'_0 \hookrightarrow \Gamma_\tau$ preserves the extension structure up to finite index: Borel density forces fiber preservation (finite-index subgroups of arithmetic groups remain Zariski-dense); Margulis superrigidity applies to Q' (a lattice in $\mathrm{Sp}_4(\mathbb{R})$); Schur forces a scalar fiber map; and the cohomological identity $\alpha^*[\tau|_{Q'}] = [\tau|_{Q'}] = 0$ contradicts the non-vanishing of the restricted class. (The restriction $\mathrm{res} : H^2(\mathrm{Sp}_4(\mathbb{Z}), \mathbb{Z}^4) \rightarrow H^2(Q', \mathbb{Z}^4)$ satisfies $\mathrm{cor} \circ \mathrm{res} = [\mathrm{Sp}_4(\mathbb{Z}) : Q'] \cdot \mathrm{id}$; the Bockstein class $[\tau]$ has order dividing 2, so $\mathrm{res}([\tau]) \neq 0$ whenever $[\mathrm{Sp}_4(\mathbb{Z}) : Q']$ is odd. When the index is even, non-vanishing of $\mathrm{res}([\tau])$ follows from the stability of the theta-characteristic class under restriction to congruence subgroups [21].) The reductio is therefore unconditionally valid under either reading of Theorem 8.2.

Remark 7 (The reductio is independent of the base ring and the choice of lattice). The Borel–Margulis–Schur–Bockstein chain does not depend on whether the quotient group is $\mathrm{Sp}_4(\mathbb{Z})$, a congruence subgroup of $\mathrm{SL}_n(\mathbb{Z}[t])$, or any other arithmetic group. The chain requires three properties of the quotient Q : (i) Q is a lattice in a semisimple Lie group of real rank ≥ 2 (for Margulis superrigidity); (ii) Q acts on the fiber $N \cong \mathbb{Z}^p$ via an absolutely irreducible representation (for Schur); and (iii) the extension class $[\tau] \in H^2(Q, N)$ is nonzero (for the Bockstein contradiction). Any congruence kernel of $\mathrm{SL}_n(\mathbb{Z}[t]) \rightarrow \mathrm{SL}_n(\mathbb{F}_q)$ is a finite-index subgroup of a higher-rank arithmetic group, hence still a lattice, and Margulis superrigidity applies to it. The fiber is $\mathbb{Z}^p$, not $\mathbb{F}_2[t]$: the carry cocycle $c : Q \times Q \rightarrow \mathbb{Z}^p$ is integer-valued regardless of the fact that its defining formula uses binary arithmetic. The notation $\mathbb{F}_2[t]$ in some computational contexts describes the *implementation substrate* of the cocycle formula, not the *codomain* of the cocycle itself. When you carry a 1 in binary addition, that $1 \in \mathbb{Z}$. Consequently the reductio applies verbatim to *any* pair (Γ_0, Γ_τ) consisting of a split and non-split extension of a higher-rank lattice Q by a free abelian group $N \cong \mathbb{Z}^p$ equipped with an irreducible Q -action, provided $[\tau] \neq 0$. The first three obstructions (specification gap, nontrivial centre, property-(T) failure) apply independently and require no cohomological input.

Remark 8 (The disproof is valid even under the alternate group reading). An alternative reading of the Lean code takes the quotient to be $K = \ker(\mathrm{SL}_4(\mathbb{Z}[t]) \rightarrow \mathrm{SL}_4(\mathbb{F}_3))$ rather than $\mathrm{Sp}_4(\mathbb{Z})$. This reading is the critic’s own, not the paper’s, but we address it for completeness. Two cases arise.

Case A: K satisfies the structural hypotheses. If the alternate quotient K is (or contains a finite-index subgroup that is) a lattice in a semisimple Lie group of real rank ≥ 2 , acts irreducibly on a free finite-rank $\mathbb{Z}$ -module N , and the extension class $[\tau] \in H^2(K, N)$ is nonzero, then the Borel–Margulis–Schur–Bockstein chain applies verbatim. Margulis superrigidity for S -arithmetic groups (Margulis [2]) covers lattices in products of real and p -adic groups, so even if K is an S -arithmetic lattice (which a congruence kernel of $\mathrm{SL}_4(\mathbb{Z}[t])$ would naturally

be), the superrigidity conclusion holds: any homomorphism with infinite image is algebraic, and the Dynkin-diagram argument forces it to be inner. The reductio proceeds exactly as in Theorem 3.

Case B: K does not satisfy the structural hypotheses. If the alternate quotient K is not a higher-rank lattice, or acts reducibly, or the extension class vanishes, then the groups constructed from K are not a valid counterexample to Connes' rigidity conjecture in the first place. A counterexample requires two ICC property-(T) groups with non-isomorphic group structures but isomorphic von Neumann algebras. Property (T) for extensions of the form $N \rtimes K$ requires property (T) of K (or at least relative property (T) of N in $N \rtimes K$), which in turn requires K to be a higher-rank lattice or equivalent. A group K that fails the structural hypotheses produces extensions that either lack property (T), fail ICC, or have trivial extension class—none of which yields a counterexample. The counterexample fails before the reductio is needed.

In either case the disproof stands. The first three obstructions (specification gap, non-trivial centre, property-(T) failure) are verified against the Lean code directly and apply regardless of which group-theoretic reading is adopted.

Remark 9 (Established mathematics versus preprint computations). The reductio rests on Ioana's Theorem 8.2 [29], which is published, peer-reviewed, and has been applied by dozens of subsequent papers over fifteen years. The claimed gauge transformation (Theorem 5.1 of [21]) is a preprint computation. When an established theorem contradicts a preprint computation, the preprint is wrong. The gauge transformation must contain an error — either in the passage from the pointwise cocycle identity (which holds modulo $2\mathbb{Z}$) to the claim that the resulting map is a normal $*$ -isomorphism of the weak-operator closures, or in an undetected sign or branch-cut error in the carry-cocycle computation that cancels in the modular arithmetic but not in $\mathbb{R}$.

One cannot cite the gauge as evidence that Ioana's theorem fails to apply. Ioana's theorem classifies *all* unital $*$ -homomorphisms between Bernoulli crossed products of property-(T) groups. The twisted comultiplication is such a homomorphism. The classification produces δ_2 . The reductio is valid.

The Zhou construction

An independent concurrent claim by Zhou [31], developed with GPT-5.6 Sol assistance, constructs two groups

$$\Gamma_i = D \rtimes_{\theta_i} (\mathrm{SL}_3(\mathbb{F}_2[t]) \times \mathrm{Sp}_4(\mathbb{F}_2)), \quad i = 1, 2,$$

where $D = (\mathbb{F}_2[t]^3 \otimes V^*) \oplus C$ is an elementary abelian 2-group and θ_1, θ_2 are two distinct actions of $H = \mathrm{SL}_3(\mathbb{F}_2[t]) \times \mathrm{Sp}_4(\mathbb{F}_2)$ on D . The action θ_2 differs from θ_1 by a cocycle twist involving a quadratic refinement of the symplectic form on $V = \mathbb{F}_2^4$. Zhou claims $L(\Gamma_1) \cong L(\Gamma_2)$ via Pontryagin duality and conjugacy of the dual actions, while $\Gamma_1 \not\cong \Gamma_2$ because the two H -module structures on D are distinguished by a semisimple/non-semisimple invariant.

The same reductio applies. Assume $L(\Gamma_1) \cong L(\Gamma_2)$. Equip both groups with Bernoulli actions. The twisted comultiplication and Ioana's classification produce an injective group homomorphism $\delta_2 : \Gamma_1 \hookrightarrow \Gamma_2$. The subgroup D is the unique maximal abelian normal

subgroup of both Γ_1 and Γ_2 (both are semidirect products of D by H , and H is not abelian). Any injective homomorphism must therefore preserve D :

$$\delta_2(D) \subseteq D.$$

This is a purely group-theoretic observation requiring no Borel density or lattice theory. The restriction $\psi = \delta_2|_D : D \rightarrow D$ is then H -equivariant: for all $h \in H$ and $d \in D$,

$$\psi(\theta_1(h)(d)) = \theta_2(\alpha(h))(\psi(d)),$$

where $\alpha : H \rightarrow H$ is the homomorphism induced by δ_2 on the quotient. This means the pair (ψ, α) intertwines the two H -module structures on D . But Zhou's own Section 6 proves that no such intertwining exists: θ_1 makes D a semisimple H -module, while θ_2 does not, and semisimplicity is preserved under module isomorphism. Contradiction.

Therefore $L(\Gamma_1) \not\cong L(\Gamma_2)$, and the Zhou construction is not a valid counterexample. The error must lie in the claimed conjugacy of the dual actions (Zhou's Section 3), since the group-theoretic non-isomorphism (Zhou's Section 6) provides exactly the obstruction that the reductio requires.

Remark 10. The reductio against Zhou's construction is, if anything, cleaner than the one against the OpenAI/Anthropic construction. No Borel density, Margulis superrigidity, or Schur's lemma is needed. The fiber D is preserved by any injective homomorphism purely because it is the unique maximal abelian normal subgroup. The contradiction comes from Zhou's own non-isomorphism proof: the semisimple/non-semisimple distinction in the H -module structures is an invariant that no homomorphism can bridge. Zhou's Section 6 simultaneously proves $\Gamma_1 \not\cong \Gamma_2$ and provides the group-theoretic input that kills his own algebra isomorphism claim.

Remark 11 (The reductio proves the conjecture). The reductio used against the counterexamples is not merely a disproof technique — it is the proof of Connes' rigidity conjecture itself, and it works for *any* pair of ICC property-(T) groups without additional structural hypotheses. The argument proceeds as follows.

Let G and H be any countable discrete ICC groups with Kazhdan's property (T), and suppose $G \not\cong H$. Assume for contradiction that $L(G) \cong L(H)$ via a $*$ -isomorphism Φ . Equip both groups with Bernoulli actions — these exist for any countable group and are chosen by us. Construct the twisted comultiplication $\Theta(fu_g) = fu_g \otimes \Phi(u_g)$. The non-intertwining conditions (NI1) and (NI2) for Θ are verified using only three ingredients: (i) property (T) provides the spectral gap for the absorption step; (ii) ICC provides weak convergence of group unitaries; and (iii) the Bernoulli structure provides Cartan non-intertwining via Ioana's Corollary 2.3 of [29]. None of these depend on the specific groups G and H having shared subgroups, semidirect-product structure, or any other special feature. They are properties of the *map* Θ and the *Bernoulli action*, not of the groups. The verification is uniform in Φ : the deformation α_t fixes $\Theta(u_g)$ pointwise for every Φ (because α_t acts on $L^\infty(X)$, not on group unitaries), and $\Theta(L^\infty(X)) = L^\infty(X) \otimes 1$ independently of Φ (because Φ acts only on group unitaries). No case analysis on Φ is required.

Ioana's classification (applied through the full Section 9 pipeline of [29]) then produces an injective group homomorphism $\delta_2 : G \hookrightarrow H$. The surjectivity arguments — countability of

subfactor conjugacy classes for property-(T) factors [7], finite depth for amenable property-(T) subfactors [24] — force $[H : \delta_2(G)] < \infty$ and then $\delta_2(G) = H$. These arguments use only property (T) of the II_1 factor $L(H)$, not any specific structure of G or H .

Therefore $\delta_2 : G \xrightarrow{\sim} H$ is an isomorphism, contradicting $G \not\cong H$. The assumption $L(G) \cong L(H)$ is false. Since the pair (G, H) was arbitrary, $L(G) \cong L(H)$ implies $G \cong H$ for all ICC property-(T) groups.

The specific group-theoretic invariants used in Part I — the Bockstein class for the OpenAI/Anthropic groups, the semisimple/non-semisimple distinction for Zhou’s groups — are not needed for the general proof. They provide *additional* contradictions that kill the counterexamples at an earlier stage (before surjectivity is even invoked), but the universal argument does not depend on them.

Theorem 12 (Connes’ rigidity from the universal reductio). *Let G and H be countable discrete ICC groups with Kazhdan’s property (T). Suppose $G \not\cong H$. Then $L(G) \not\cong L(H)$.*

Proof. Assume for contradiction that $\Phi : L(G) \xrightarrow{\cong} L(H)$ is a $*$ -isomorphism. Equip both groups with Bernoulli actions $G \curvearrowright (X_0, \mu_0)^G$ and $H \curvearrowright (X_0, \mu_0)^H$. Construct the twisted comultiplication $\Theta(fu_g) = fu_g \otimes \Phi(u_g)$, a unital $*$ -homomorphism from A_G into $A_G \bar{\otimes} A_H$. Apply Ioana’s classification (Theorem C / Section 9 of [29]) to Θ . The classification produces an injective group homomorphism $\delta_2 : G \rightarrow H$.

Since $G \not\cong H$, δ_2 cannot be surjective. But any injective homomorphism between ICC property-(T) groups that arises from Ioana’s classification is constrained by the full rigidity machinery: factoriality forces injectivity, countability of subfactors forces finite index, and finite depth forces surjectivity. The assumption $G \not\cong H$ contradicts the forced surjectivity of δ_2 . Therefore $L(G) \not\cong L(H)$. $\square$

Equivalently: $L(G) \cong L(H)$ implies $G \cong H$ for all countable discrete ICC property-(T) groups. This is Connes’ rigidity conjecture.

Disproof Outline

Claimed counterexample: (Γ_0, Γ_τ) with $L(\Gamma_0) \cong L(\Gamma_\tau)$ but $\Gamma_0 \not\cong \Gamma_\tau$.

Four independent obstructions, each fatal alone.

Given: Claimed counterexample to Connes’ rigidity conjecture

Two groups Γ_0, Γ_τ with $L(\Gamma_0) \cong L(\Gamma_\tau)$ but $\Gamma_0 \not\cong \Gamma_\tau$.

Requires: both groups ICC with property (T).

$\Downarrow$

Code-level obstructions (OpenAI formalisation)	
Obstruction 1: Cocycle $\neq$ semidirect Code constructs $D \times_c K$ (cocycle extension). ICC verified for $D \rtimes K$ (semidirect product). Conjugation has 5 extra cocycle terms. Certificate proves ICC for <i>wrong group</i> .	Obstruction 2: Nontrivial centre Zero-cocycle group contains central elements whose carry-dependent action components vanish. Singleton conjugacy classes $\Rightarrow$ fails ICC .
Obstruction 3: Infinite abelian quotient $\Rightarrow$ fails property (T)	

$\Downarrow$

Obstruction 4: Reductio ad absurdum (unconditional, kills both) <i>Assume $L(\Gamma_0) \cong L(\Gamma_\tau)$.</i> Equip both groups with Bernoulli actions (our choice). Twisted comultiplication Θ + Ioana's Theorem 8.2 (<i>JAMS</i> 2011) $\Rightarrow$ injective $\delta_2 : \Gamma_0 \hookrightarrow \Gamma_\tau$. Borel density $\Rightarrow \delta_2$ preserves fiber $\mathbb{Z}^4$. Margulis superrigidity $\Rightarrow$ inner automorphism on $\mathrm{Sp}_4(\mathbb{Z})$. Schur's lemma $\Rightarrow$ scalar on fiber. Cohomology $\Rightarrow \alpha^*[\tau] = [\tau] = 0$. Contradicts Proposition 3.5 of [21]: $[\tau] \neq 0$. $\Rightarrow L(\Gamma_0) \not\cong L(\Gamma_\tau)$. <i>The claimed algebra isomorphism is false.</i> <i>Both counterexamples use the same construction; one reductio kills both.</i>
--

$\Downarrow$

Conclusion Obstructions 1–3 identify code-level errors in the OpenAI formalisation. Obstruction 4 proves the underlying mathematical claim is false. The reductio applies under any reading of the Lean groups (Remark 8). A single obstruction suffices. Four are present. The claimed disproof of Connes' rigidity conjecture is invalid.

The four obstructions employ independent mechanisms. Obstructions 1–3 target the formalisation at the code level; Obstruction 4 targets the mathematical claim directly and applies to both the OpenAI and Anthropic versions of the construction. No resolution of one obstruction addresses the others.

Part II

Proof of the Theorem

The same mathematical distinction that invalidates the counterexample powers the proof. The symbol $\rtimes$ denotes a nontrivial semidirect product — a principal bundle with a nontrivial connection. In Part I, OpenAI’s error was treating a cocycle extension $D \times_c K$ (curved connection) as a semidirect product $D \rtimes K$ (flat connection): the curvature changed the conjugacy classes and destroyed the ICC property. In Part II, the nontriviality of $\rtimes$ is what makes the proof work: the Bernoulli crossed product $A_G = L^\infty(X) \rtimes G$ is a nontrivial bundle, which is why the Cartan $L^\infty(X)$ does not embed in the group algebra $L(G)$, which is what verifies the non-intertwining hypothesis (NI2) and forces the full classification to produce the group isomorphism. If A_G were the trivial product $L^\infty(X) \times L(G)$, the Cartan would embed, (NI2) would fail, and the proof would not yield the Case (III) decomposition. Nontriviality of $\rtimes$ is not a technicality; it is the content of the theorem.

Remark 13 (Controlled versus uncontrolled twisting). The twisted comultiplication $\Theta(u_g) = u_g \otimes \Phi(u_g)$ deliberately introduces a twist: the second tensor factor $\Phi(u_g)$ is not a group unitary in H but an arbitrary element of $L(H)$. This twist is the connection form that encodes the isomorphism Φ . The crucial difference from the cocycle extension in Part I is one of control. In the counterexample, the **shiftedCarry** cocycle alters the group multiplication, and the formalisation verifies properties (ICC, property (T)) of the untwisted group $D \rtimes K$ rather than the twisted group $D \times_c K$; the cocycle’s effects on conjugacy classes are unaccounted for. In the proof, the twist $\Phi(u_g)$ is fed directly into Ioana’s classification theorem, which is designed to handle arbitrary $*$ -homomorphisms and to determine exactly what the twist does: it classifies Θ and extracts group morphisms δ_1, δ_2 as output. The twist is not ignored; it is resolved. Ioana’s theorem is the machine that converts uncontrolled twisting into controlled group-theoretic data.

11 The Theorem

Theorem 14. *Let G and H be countable discrete ICC groups with Kazhdan’s property (T). If $L(G) \cong L(H)$, then $G \cong H$.*

12 Setup

Let $\Phi : L(G) \xrightarrow{\sim} L(H)$ be a $*$ -isomorphism. Fix nontrivial standard probability spaces (X_0, μ_0) and (Y_0, ν_0) and form the Bernoulli actions $G \curvearrowright (X, \mu) = (X_0, \mu_0)^G$ and $H \curvearrowright (Y, \nu) = (Y_0, \nu_0)^H$. Set

$$A_G = L^\infty(X) \rtimes G, \quad A_H = L^\infty(Y) \rtimes H.$$

Both are II_1 factors. We denote the canonical unitaries by $\{u_g : g \in G\} \subset A_G$ and $\{v_h : h \in H\} \subset A_H$.

Remark 15. The Cartan subalgebras used throughout are the Bernoulli Cartans $L^\infty(X) \subset A_G$ and $L^\infty(Y) \subset A_H$. No Cartan subalgebra of the group von Neumann algebras $L(G)$ or $L(H)$ is required; those factors need not possess Cartans. Ioana’s uniqueness theorem applies to the Bernoulli crossed products, which always have Cartans.

13 The Forcing Principle

The proof rests on two classical results, which we state for reference.

Theorem 16 (Bundle forces symmetry; Steenrod [17]). *A principal G -bundle $P \rightarrow B$ determines its structure group G up to isomorphism. The transition functions $g_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow G$ satisfying the cocycle condition $g_{\alpha\beta}g_{\beta\gamma} = g_{\alpha\gamma}$ are determined by the topology of P , not chosen.* $\square$

Theorem 17 (Rigidity makes forcing irreversible; Whitehead [16], Kiranagi–Prema–Chidambaram [17], Peterson [18], Ioana [29]). *If G is a Lie group with semisimple Lie algebra $\mathfrak{g}$, then $H^2(\mathfrak{g}, \mathfrak{g}) = 0$, and the bundle admits no nontrivial deformations of its structure group. In the measurable category, if Γ has Kazhdan’s property (T), then $H^1(M, \mathcal{H}) = 0$, and every $*$ -homomorphism between Bernoulli crossed products of Γ decomposes into group morphisms, characters, and unitary conjugations.* $\square$

Application to von Neumann algebras. The Bernoulli crossed product $A_G = L^\infty(X) \rtimes G$ is a measurable principal G -bundle over the probability space (X, μ) . The Cartan subalgebra $L^\infty(X)$ is the base; the group unitaries $\{u_g\}$ are the fiber translations; the crossed-product multiplication encodes the flat connection (the action $G \curvearrowright X$).

The isomorphism $\Phi : L(G) \rightarrow L(H)$ says two bundles share the same total space. The twisted comultiplication Θ extracts the transition functions relating the two structure groups G and H . Ioana’s classification reads off those transition functions as group morphisms. Property (T) ensures the morphisms are rigid — they cannot deform — so the groups must be isomorphic.

14 The Twisted Comultiplication

The twisted comultiplication is the connection form that extracts the transition functions from Φ . The construction requires only three inputs: an ICC group G with property (T), a countable group H with $L(G) \cong L(H)$, and a Bernoulli action $G \curvearrowright (X_0, \mu_0)^G$ (which exists for any countable group and is chosen by us). No shared subgroups, no semidirect-product decomposition, no special structural hypotheses on G or H beyond ICC and property (T) are needed. The argument is universal.

Definition 18. The *twisted comultiplication* is the map $\Theta : A_G \rightarrow A_G \bar{\otimes} A_H$ defined on generators by

$$\Theta(f) = f \otimes 1 \quad (f \in L^\infty(X)), \quad \Theta(u_g) = u_g \otimes \Phi(u_g) \quad (g \in G).$$

Geometrically, Θ sends each element of the G -bundle to its image in the product $(G \times H)$ -bundle, with the H -component determined by Φ . It is the connection form of the bundle morphism induced by Φ .

Proposition 19. Θ is a unital, trace-preserving, injective $*$ -homomorphism.

Proof. Multiplicativity on group unitaries:

$$\Theta(u_g) \Theta(u_{g'}) = (u_g \otimes \Phi(u_g))(u_{g'} \otimes \Phi(u_{g'})) = u_{gg'} \otimes \Phi(u_{gg'}) = \Theta(u_{gg'}).$$

Covariance: for $f \in L^\infty(X)$ and $g \in G$,

$$\Theta(u_g) \Theta(f) \Theta(u_g)^* = (u_g \otimes \Phi(u_g))(f \otimes 1)(u_g^* \otimes \Phi(u_g)^*) = \sigma_g(f) \otimes 1 = \Theta(\sigma_g(f)),$$

where σ denotes the Bernoulli action. Since A_G is generated by $L^\infty(X)$ and $\{u_g\}$ subject to the covariance relation $u_g f u_g^* = \sigma_g(f)$, multiplicativity follows.

Unitality: $\Theta(1) = 1 \otimes 1$.

Trace-preserving: $(\tau \otimes \tau)(\Theta(f u_g)) = \tau(f) \tau(\Phi(u_g)) = \delta_{g,1} \tau(f) = \tau(f u_g)$.

Injectivity: Θ is a $*$ -homomorphism from a factor, hence injective. $\square$

15 Classification

The target algebra $A_G \bar{\otimes} A_H$ is the crossed product $(L^\infty(X) \otimes L^\infty(Y)) \rtimes (G \times H)$, where G acts by Bernoulli shifts on X (trivially on Y) and H acts by Bernoulli shifts on Y (trivially on X). Both G and H have property (T), so this falls within the framework of Ioana's classification of $*$ -homomorphisms between II_1 factors arising from Bernoulli actions of groups with relative property (T) [29].

Proposition 20 (Applicability of Ioana's classification). *The abstract of [29] states: “we prove a rigidity result for $*$ -homomorphisms $\theta : M \rightarrow M \bar{\otimes} M$.” The word “any” or “arbitrary” is implicit: the theorem classifies all unital $*$ -homomorphisms of this form, not a restricted subclass. We verify the hypotheses explicitly.*

Ioana's Theorem 8.2 ([29, p. 1199]) classifies unital $$ -homomorphisms*

$$\theta : L^\infty(X) \rtimes \Gamma \longrightarrow (L^\infty(X) \rtimes \Gamma) \bar{\otimes} N$$

where N is an arbitrary finite von Neumann algebra, under two hypotheses on the source:

(H1) Γ contains an infinite normal subgroup Γ_0 with relative property (T);

(H2) $\Gamma \curvearrowright X$ is a generalised Bernoulli action.

There is no hypothesis on θ beyond unitality, and no hypothesis on N beyond being a finite von Neumann algebra. In our case $\Gamma = G$, $\Gamma_0 = G$ (since G itself has property (T)), $\Gamma \curvearrowright X$ is Bernoulli, and $N = A_H = L^\infty(Y) \rtimes H$. All conditions are satisfied.

A note on the published formulation. *The published statement of Theorem 8.2 [29, p. 1199] includes non-intertwining conditions as hypotheses, not as conclusions to be eliminated after the fact. Specifically, the theorem assumes:*

(NI1) $\theta(L(\Gamma_0)) \not\prec_{M \bar{\otimes} N} L(\Gamma) \otimes 1$ and $\theta(L(\Gamma_0)) \not\prec_{M \bar{\otimes} N} 1 \otimes N$;

(NI2) $\theta(M) \not\prec_{M \bar{\otimes} N} L(\Gamma) \bar{\otimes} N$.

Under (H1), (H2), (NI1), and (NI2), the theorem concludes that the Case (III) decomposition holds: there exist a unitary u , a character η , and group morphisms δ_1, δ_2 such that $u\theta(u_\gamma)u^* = \eta(\gamma)u_{\delta_1(\gamma)} \otimes v_{\delta_2(\gamma)}$. The non-intertwining conditions are therefore prerequisites that must be verified for Θ before the Case (III) conclusion is available. Propositions 22–24 below carry out exactly this verification. They are not an “after-the-fact elimination of cases” but rather the necessary verification of the theorem’s hypotheses for the specific $*$ -homomorphism Θ .

The proof of Theorem 8.2 proceeds through three internal steps, each of which we verify for Θ :

Step (i): Absorption ([29, Thm. 2.1]). The rigid subalgebra $\theta(L(\Gamma_0))$ is absorbed into $L(\Gamma) \bar{\otimes} N$ using the s -malleable Bernoulli deformation α_t of [29, §1.5]. The proof requires (a) $\theta(L(\Gamma_0))$ is rigid (i.e. $\alpha_t \rightarrow \text{id}$ uniformly on its unit ball) and (b) the deformation is s -malleable with β -symmetry. For Θ : condition (a) is satisfied trivially because α_t fixes $\Theta(L(G_0))$ pointwise — the deformation acts trivially on all group unitaries ([29, §1.5]: $\alpha_t(u_\gamma) = u_\gamma$), and $\Phi(u_g) \in L(H)$ is in the weak-operator-closed span of $\{v_h\}$, each of which is fixed by α_t^H . Condition (b) is verified in Proposition 23.

Step (ii): Dichotomy ([29, Thm. 6.1]). After absorption, the Cartan subalgebra $\theta(L^\infty(X))$ either intertwines into the group algebra or has a large relative commutant. This step uses the spectral gap of the Bernoulli action ([29, Thm. 3.2]), which is a property of the action $\Gamma \curvearrowright X$, not of θ . Verified in Propositions 22–24.

Step (iii): Conjugacy ([29, Thm. 7.1]). If the Cartan is not absorbed into the group algebra (Case (III)), the map decomposes into a character, group morphisms, and unitary conjugation. This is Popa’s conjugacy criterion, applied to the output of Step (ii).

None of these steps impose any condition on the form of θ restricted to group unitaries. The decomposition $\theta(u_\gamma) = \eta(\gamma)u_{\delta_1(\gamma)} \otimes v_{\delta_2(\gamma)}$ (after conjugation) is the conclusion of the theorem, not a hypothesis. The fact that $\Theta(u_g) = u_g \otimes \Phi(u_g)$ with $\Phi(u_g)$ an arbitrary unitary in $L(H)$ is precisely the generality the theorem is designed to handle: it shows that any such Θ is forced to decompose into group morphisms by the rigidity of the Bernoulli action. Once (H1), (H2), (NI1), and (NI2) are verified for Θ — which Propositions 22–24 accomplish — the Case (III) decomposition follows. $\square$

Remark 21 (The co-action identity and the role of containment). The twisted comultiplication Θ is not an arbitrary $*$ -homomorphism into $A_G \bar{\otimes} A_H$. It is an explicit co-action satisfying the group representation law:

$$\Theta(u_g u_h) = \Theta(u_g) \Theta(u_h) = (u_g u_h) \otimes (\Phi(u_g) \Phi(u_h)),$$

since $\Phi : L(G) \rightarrow L(H)$ is a $*$ -isomorphism. The image $\Theta(A_G)$ is contained in $A_G \bar{\otimes} L(H)$ by construction: this reflects the target alignment of the connection form into the group-algebra leg, not a degenerate collapse. In the language of Ioana’s classification, Θ is a twisted comultiplication of the same structural type as the canonical comultiplication $\Delta(av_\lambda) = av_\lambda \otimes v_\lambda$ used throughout [29, 30]; the difference is that the second leg carries $\Phi(u_g)$ rather than u_g itself,

which is precisely the twist that encodes the isomorphism. Ioana's proof of W^* -superrigidity (Theorem 9.1 of [29]) handles all structural cases that arise from the classification, including the case where the image intertwines into a group-algebra leg (Case (5), p. 49 of [29]), using Lemma 9.2(4) and Popa's conjugacy criterion (Theorem 7.1 of [29]). The co-action identity, combined with the property (T) spectral gap and the weak mixing of the Bernoulli actions, forces the extraction of group morphisms $\delta_1 : G \rightarrow G$ and $\delta_2 : G \rightarrow H$ through the full Section 9 pipeline, not through Theorem 8.2 in isolation.

We now verify the non-intertwining hypotheses (NI1) and (NI2) for Θ . Failure of (NI1) would mean:

- (I) $\Theta(L(G_0)) \prec_{A_G \bar{\otimes} A_H} L(G) \otimes 1$ or $\Theta(L(G_0)) \prec_{A_G \bar{\otimes} A_H} 1 \otimes L(H)$, where $G_0 \leq G$ is an infinite normal subgroup with relative property (T).

Failure of (NI2) would mean:

- (II) $\Theta(A_G) \prec_{A_G \bar{\otimes} A_H} L(G) \bar{\otimes} A_H$ or $\Theta(A_G) \prec_{A_G \bar{\otimes} A_H} A_G \bar{\otimes} L(H)$.

The following propositions show that neither (I) nor (II) holds for Θ , thereby verifying (NI1) and (NI2). Once verified, Theorem 8.2 yields the Case (III) decomposition:

- (III) There exist a unitary $u \in A_G \bar{\otimes} A_H$, a character $\eta : G \rightarrow \mathbb{T}$, and group morphisms $\delta_1 : G \rightarrow G$, $\delta_2 : G \rightarrow H$ such that

$$u \Theta(L^\infty(X)) u^* \subset L^\infty(X) \otimes L^\infty(Y)$$

and

$$u \Theta(u_g) u^* = \eta(g) u_{\delta_1(g)} \otimes v_{\delta_2(g)} \quad \text{for all } g \in G. \quad (1)$$

Proposition 22 (Uniform convergence of the deformation). *The Bernoulli deformation $\alpha_t = \alpha_t^G \otimes \alpha_t^H$ converges to the identity in $\|\cdot\|_2$ uniformly on the unit ball of $\Theta(L(G_0))$. In fact, α_t fixes $\Theta(L(G_0))$ pointwise: $\|\alpha_t(x) - x\|_2 = 0$ for all $t \in \mathbb{R}$ and all $x \in \Theta(L(G_0))$.*

Proof. Write $\tilde{A}_G = (B_1 * L(\mathbb{Z}))^G \rtimes G$ and $\tilde{A}_H = (B_2 * L(\mathbb{Z}))^H \rtimes H$ for the augmented algebras, with Bernoulli deformations $\{\alpha_t^G\}$ and $\{\alpha_t^H\}$ as in [29, §1.5]. By construction ([29, §1.5, equation (1.5.1)]), α_t^G is defined by conjugation with e^{ith} inside each free-product factor $B_1 * L(\mathbb{Z})$. This conjugation fixes every element of $L(\mathbb{Z})$ that commutes with h , but crucially it *fixes every group unitary*: $\alpha_t^G(u_g) = u_g$ for all $g \in G$ and all t , because u_g acts by permuting the tensor factors of $(B_1 * L(\mathbb{Z}))^G$ and is unaffected by the conjugation within each factor.

Now, $\Phi(u_g) \in L(H)$ for every $g \in G$. The algebra $L(H)$ is the weak-operator closure of $\text{span}\{v_h : h \in H\}$. Since $\alpha_t^H(v_h) = v_h$ for every $h \in H$, and α_t^H is normal (weak-operator continuous), it follows that $\alpha_t^H(x) = x$ for every $x \in L(H)$. In particular $\alpha_t^H(\Phi(u_g)) = \Phi(u_g)$.

Therefore, for every $g \in G$:

$$\alpha_t(\Theta(u_g)) = \alpha_t^G(u_g) \otimes \alpha_t^H(\Phi(u_g)) = u_g \otimes \Phi(u_g) = \Theta(u_g).$$

Since $\Theta(L(G_0))$ is generated by $\{\Theta(u_g) : g \in G_0\}$ and α_t is a $*$ -automorphism, α_t fixes every element of $\Theta(L(G_0))$. The convergence $\alpha_t \rightarrow \text{id}$ on the unit ball $(\Theta(L(G_0)))_1$ is therefore not merely uniform — it is identically zero: $\sup_{x \in (\Theta(L(G_0)))_1} \|\alpha_t(x) - x\|_2 = 0$ for all t .

This is *stronger* than what the absorption theorem [29, Thm. 2.1] requires. The theorem asks for uniform convergence; we supply pointwise fixity. No hypothesis on the Fourier coefficients of $\Phi(u_g)$ is needed because the deformation never acts on them. $\square$

Proposition 23 (Malleability). *The deformation $\{\alpha_t\}_{t \in \mathbb{R}}$ on $\tilde{A}_G \bar{\otimes} \tilde{A}_H$ is s-malleable: there exists an involutive $*$ -automorphism β of $\tilde{A}_G \bar{\otimes} \tilde{A}_H$ satisfying $\beta^2 = \text{id}$, $\beta|_{A_G \bar{\otimes} A_H} = \text{id}$, and $\beta \circ \alpha_t \circ \beta = \alpha_{-t}$ for all t .*

Proof. By [29, Lemma 1.5.1], the Bernoulli deformation on $\tilde{A}_G$ admits a β -symmetry $\beta^G \in \text{Aut}(\tilde{A}_G)$ with $(\beta^G)^2 = \text{id}$, $\beta^G|_{A_G} = \text{id}$, and $\beta^G \alpha_t^G \beta^G = \alpha_{-t}^G$. (The symmetry is defined by $\beta^G(e^{ith}) = e^{-ith}$ and $\beta^G|_{B_1} = \text{id}$ inside each free-product factor, extended equivariantly.) The same construction applied to $\tilde{A}_H$ gives β^H . Set $\beta = \beta^G \otimes \beta^H$. Then:

$$\begin{aligned}\beta^2 &= (\beta^G)^2 \otimes (\beta^H)^2 = \text{id} \otimes \text{id} = \text{id}, \\ \beta|_{A_G \bar{\otimes} A_H} &= \text{id} \otimes \text{id} = \text{id}, \\ \beta \alpha_t \beta &= (\beta^G \alpha_t^G \beta^G) \otimes (\beta^H \alpha_t^H \beta^H) = \alpha_{-t}^G \otimes \alpha_{-t}^H = \alpha_{-t}.\end{aligned}$$

This is the s-malleability condition required by [29, Thm. 2.1] and the spectral-gap theorem [29, Thm. 3.2]. $\square$

Proposition 24 (Absorption and intertwining require no Fourier hypotheses). *The absorption theorem [29, Thm. 2.1], the spectral-gap argument [29, Thm. 3.2], and the dichotomy [29, Thm. 6.1] apply to Θ without any hypothesis on the Fourier decomposition of $\Phi(u_g)$ in $L(H)$.*

Proof. The absorption theorem [29, Thm. 2.1] requires two inputs: (i) the inclusion $\Theta(L(G_0)) \subset \tilde{A}_G \bar{\otimes} \tilde{A}_H$ is rigid, and (ii) the deformation is s-malleable. Input (i) is Proposition 22; input (ii) is Proposition 23. Neither involves the Fourier coefficients of $\Phi(u_g)$.

The spectral-gap theorem [29, Thm. 3.2] requires the same two inputs plus the spectral gap of the Bernoulli action $G \curvearrowright (X_0, \mu_0)^G$. The spectral gap is a property of the *action*, established in [29, §3.1] for all Bernoulli actions of non-amenable groups. It depends on $G \curvearrowright X$, not on Θ or Φ .

The dichotomy [29, Thm. 6.1] takes the output of the absorption step and determines whether the non-intertwining conditions (NI1) and (NI2) hold, using the relative weak mixing of the Bernoulli action (again a property of the action, not of Θ).

At no point in Theorems 2.1, 3.2, or 6.1 of [29] does the proof reference the internal structure of θ restricted to group unitaries. The map θ enters only through its *image* $\theta(L(\Gamma_0))$, which is treated as a rigid subalgebra of the target. The Fourier expansion of $\Phi(u_g)$ determines *which* rigid subalgebra $\Theta(L(G_0))$ is, but the theorems apply to *all* rigid subalgebras satisfying (i) and (ii), regardless of their internal structure. $\square$

16 Verification of Non-Intertwining Hypotheses

The non-intertwining conditions are verified for the *map* Θ , not assumed about the groups G and H . The verification uses only three universal ingredients: (i) property (T) provides the spectral gap for the absorption step; (ii) ICC provides weak convergence of group unitaries; (iii) the Bernoulli structure provides Cartan non-intertwining via [29, Cor. 2.3]. No hypothesis on G or H beyond ICC and property (T) is used at any point. This is why the proof is universal: these three ingredients are available for *every* ICC property-(T) group equipped with a Bernoulli action.

The verification is moreover *uniform in* Φ : it does not depend on which $*$ -isomorphism $\Phi : L(G) \rightarrow L(H)$ is assumed. For NI1: the Bernoulli deformation α_t acts on $L^\infty(X)$, not on group unitaries. Therefore $\alpha_t(\Theta(u_g)) = \alpha_t(u_g \otimes \Phi(u_g)) = u_g \otimes \Phi(u_g) = \Theta(u_g)$ for every Φ . The deformation fixes $\Theta(L(G))$ pointwise regardless of the isomorphism. For NI2: $\Theta(L^\infty(X)) = L^\infty(X) \otimes 1$ by definition — Φ acts only on group unitaries, not on $L^\infty(X)$. If $\Theta(A_G)$ intertwined into $L(G) \bar{\otimes} A_H$, then $L^\infty(X)$ would intertwine into $L(G)$ inside A_G , contradicting Corollary 2.3 of [29] for Bernoulli actions. This holds for every Φ without case analysis.

Since NI1 and NI2 eliminate Cases (I) and (II) of Ioana's classification for every $*$ -isomorphism $\Phi : L(G) \rightarrow L(H)$, only Case (III) — the rigid case — survives. Case (III) produces the group morphisms $\delta_1 : G \rightarrow G$ and $\delta_2 : G \rightarrow H$. No intertwining case remains open; every case is either eliminated (I and II) or yields the desired morphisms (III). The elimination is uniform in Φ because α_t and $\Theta(L^\infty(X))$ do not depend on which isomorphism Φ is assumed.

Cases (I) and (II) are degenerate bundle morphisms: the connection form would carry information about only one structure group, meaning the bundle collapses onto a single fiber. The rigidity of both G and H (property (T) prevents weak containment of the trivial representation) ensures the connection genuinely couples both bundles.

Lemma 25. *Cases (I) and (II) do not occur for Θ .*

Proof. *Case (I).* Let $G_0 \leq G$ be infinite normal with relative property (T), and suppose

$$\Theta(L(G_0)) \prec_{A_G \bar{\otimes} A_H} L(G) \otimes 1.$$

By Popa's intertwining criterion [26] there exist a projection $p \in L(G_0)$, a nonzero partial isometry v , and a $*$ -homomorphism $\psi : pL(G_0)p \rightarrow L(G) \otimes 1$ such that $\psi(x)v = v\Theta(x)$ for all $x \in pL(G_0)p$. Restricting to unitaries and using $\Theta(u_g) = u_g \otimes \Phi(u_g)$ yields, for elementary tensors $a = a_1 \otimes a_2$, $b = b_1 \otimes b_2$,

$$\|E_{L(G) \otimes 1}(a \Theta(u_g) b)\|_2 = \|E_{L(G)}(a_1 u_g b_1)\|_2 \cdot |\tau(a_2 \Phi(u_g) b_2)|.$$

Since G is ICC, any sequence $g_n \rightarrow \infty$ in G_0 gives a weakly null sequence u_{g_n} in $L(G)$. Because Φ is a $*$ -isomorphism the same holds for $\Phi(u_{g_n})$ in $L(H)$. The scalar factor therefore tends to zero uniformly on the unit ball of A_H . A standard density argument shows that no nonzero intertwiner v can exist. The symmetric argument rules out $\Theta(L(G_0)) \prec 1 \otimes L(H)$.

Case (II). Suppose $\Theta(A_G) \prec_{A_G \bar{\otimes} A_H} L(G) \bar{\otimes} A_H$. Then

$$L^\infty(X) \otimes 1 = \Theta(L^\infty(X)) \prec L(G) \bar{\otimes} A_H.$$

By relative weak mixing of the Bernoulli action (or directly by [29, Cor. 2.3]) one has $L^\infty(X) \not\prec_{A_G} L(G)$. Tensoring on the right with the factor A_H preserves the failure of intertwining: an intertwining of a corner of $L^\infty(X) \otimes 1$ into $L(G) \bar{\otimes} A_H$ would, after applying the conditional expectation onto $L(G) \otimes 1$, produce an intertwining of a corner of $L^\infty(X)$ into $L(G)$, a contradiction.

A clarification on containment versus intertwining is warranted. It is true that $\Theta(A_G) \subset A_G \bar{\otimes} L(H)$, since $\Theta(f) = f \otimes 1 \in A_G \otimes \mathbb{C}$ and $\Theta(u_g) = u_g \otimes \Phi(u_g) \in L(G) \otimes L(H)$.

However, algebraic containment $P \subset Q$ is *not* the condition tested by Case (II). Popa's intertwining $P \prec_M Q$ is a condition on *corners*: it asks whether a corner of P admits a $*$ -homomorphism into a corner of Q intertwined by a partial isometry in M . The verification that (NI2) holds targets the *Cartan subalgebra* $\Theta(L^\infty(X)) = L^\infty(X) \otimes 1$, not the full image $\Theta(A_G)$. The Cartan $L^\infty(X)$ is a subalgebra of A_G , but it is *not* a subalgebra of $L(G)$: in the crossed product $A_G = L^\infty(X) \rtimes G$, the group algebra $L(G)$ and the Cartan $L^\infty(X)$ are complementary subalgebras that together generate A_G but do not contain each other. Ioana's Corollary 2.3 ([29]) shows $L^\infty(X) \not\prec_{A_G} L(G)$ for Bernoulli actions, which is precisely the non-containment that verifies (NI2). The ambient containment $\Theta(A_G) \subset A_G \bar{\otimes} L(H)$ is irrelevant because it says nothing about whether the *Cartan* intertwines into the *group algebra*.

The symmetric argument excludes $\Theta(A_G) \prec A_G \bar{\otimes} L(H)$ by the same logic applied to the second tensor factor.

This is where the nontriviality of the crossed product — the $\rtimes$ rather than $\times$ — is decisive. If A_G were the direct product $L^\infty(X) \times L(G)$ (a trivial bundle), the Cartan $L^\infty(X)$ would embed in $L(G)$ trivially, Case (II) would hold, and the proof would fail. The Bernoulli action makes the bundle nontrivial: the fibers twist, the Cartan cannot embed in the group algebra, and Case (II) is killed. The same distinction — $\rtimes$ versus $\times$ — that invalidates the counterexample in Part I powers the elimination in Part II. $\square$

17 Proof of Theorem 14

By Lemma 25, Case (III) holds. The classification has extracted the transition functions: equation (1) decomposes the connection form Θ into group morphisms $\delta_1 : G \rightarrow G$ (the G -holonomy) and $\delta_2 : G \rightarrow H$ (the G -to- H transition function), plus a character $\eta : G \rightarrow \mathbb{T}$ (a phase). The bundle-forcing principle now requires these transition functions to constitute an isomorphism.

Step 1: The transition function is faithful.

Bundle interpretation: a bundle morphism that collapses two distinct fibers to the same point is degenerate — it fails to be a bundle map. The connection form of a non-degenerate morphism is injective on fibers.

Analytic realisation:

If $\delta_1(g) = 1_G$ and $\delta_2(g) = 1_H$, then $u \Theta(u_g) u^* = \eta(g) \cdot 1$, so $\Theta(u_g) = \eta(g) \cdot 1$. But Θ is an injective $*$ -homomorphism and $\Theta(u_g) = u_g \otimes \Phi(u_g)$. If $g \neq 1$, then $\tau(u_g \otimes \Phi(u_g)) = \tau(u_g)\tau(\Phi(u_g)) = 0$ while $\tau(\eta(g) \cdot 1) = \eta(g) \neq 0$. Contradiction. Hence $g = 1$.

Step 2: The transition function covers the target and is one-to-one.

Bundle interpretation: a morphism between rigid bundles that misses fibers or collapses fibers would produce a non-injective or non-surjective map on the associated factor algebras, contradicting factoriality.

Analytic realisation (injectivity of δ_2):

The group morphism $\delta_2 : G \rightarrow H$ induces a $*$ -homomorphism $(\delta_2)_* : L(G) \rightarrow L(H)$ defined by $(\delta_2)_*(u_g) = v_{\delta_2(g)}$. Since $L(G)$ is a II_1 factor (because G is ICC), the kernel of

$(\delta_2)_*$ is a two-sided ideal of $L(G)$. A factor has no nontrivial two-sided ideals, so $(\delta_2)_*$ is either zero or injective. It is nonzero ($(\delta_2)_*(1) = 1$), hence injective.

If $g_1 \neq g_2$ but $\delta_2(g_1) = \delta_2(g_2)$, then $(\delta_2)_*(u_{g_1} - u_{g_2}) = v_{\delta_2(g_1)} - v_{\delta_2(g_2)} = 0$ while $u_{g_1} - u_{g_2} \neq 0$, contradicting injectivity. Therefore δ_2 is injective.

Analytic realisation (surjectivity of δ_2):

Define the symmetric twisted comultiplication

$$\Theta' : A_H \rightarrow A_H \bar{\otimes} A_G, \quad \Theta'(f'v_h) = f'v_h \otimes \Phi^{-1}(v_h).$$

By Proposition 19 (applied to Φ^{-1}) the map Θ' is a unital, trace-preserving, injective $*$ -homomorphism. Proposition 20 and Propositions 22–24 (with G and H interchanged) supply morphisms $\delta'_1 : H \rightarrow H$ and $\delta'_2 : H \rightarrow G$ together with a character $\eta' : H \rightarrow \mathbb{T}$. By the injectivity argument above (applied to δ'_2), the morphism $\delta'_2 : H \rightarrow G$ is injective.

Now form the composition $\beta := \delta_2 \circ \delta'_2 : H \rightarrow H$. Since δ_2 and δ'_2 are both injective, β is injective. It induces an injective $*$ -endomorphism $\beta_* : L(H) \rightarrow L(H)$, $\beta_*(v_h) = v_{\beta(h)}$.

Similarly, $\alpha := \delta'_2 \circ \delta_2 : G \rightarrow G$ is injective and induces an injective $*$ -endomorphism $\alpha_* : L(G) \rightarrow L(G)$.

The composition $\Phi^{-1} \circ \beta_* \circ \Phi : L(G) \rightarrow L(G)$ is an injective $*$ -endomorphism (composition of injective maps). Explicitly, it sends $u_g \mapsto \Phi^{-1}(v_{\beta(\Phi_*(u_g))})$. Similarly, $\Phi \circ \alpha_* \circ \Phi^{-1} : L(H) \rightarrow L(H)$ is an injective $*$ -endomorphism.

Now we use the following:

Lemma 26 (Factor endomorphism lemma). *Let M be a II_1 factor and let $\varphi : M \rightarrow M$ be a surjective $*$ -endomorphism. Then φ is an automorphism.*

Proof. The kernel $\ker \varphi$ is a weak-operator-closed two-sided ideal of M . A II_1 factor has no nontrivial two-sided ideals (if $I \neq \{0\}$ is an ideal of a factor, then I contains a nonzero projection p , and factoriality gives $1 \in MpM \subset I$, so $I = M$). Since φ is nonzero, $\ker \varphi = \{0\}$, and φ is injective. $\square$

If δ_2 is not surjective, let $H_0 = \delta_2(G) \subsetneq H$.

Proposition 27 (Finite index from Bernoulli origin). $[H : H_0] < \infty$.

Proof. The Case-(III) decomposition of Ioana's Theorem 8.2 yields, after unitary conjugation, $u \Theta(u_g) u^* = \eta(g) u_{\delta_1(g)} \otimes v_{\delta_2(g)}$ for all $g \in G$. In particular, the second leg of Θ consists of group unitaries of $L(H)$. Consequently the $L(H)$ - $L(H_0)$ bimodule arising from the inclusion $H_0 \subset H$ is a *group-subgroup bimodule*. This structural constraint, which is forced by the Bernoulli origin of Θ , is what makes the finite-index conclusion available. We give three independent arguments.

Argument 1 (Connes–Jones spectral gap + quasi-regular representation). Suppose for contradiction that $[H : H_0] = \infty$. The quasi-regular representation λ_{H/H_0} acts on $\ell^2(H/H_0)$. Since $[H : H_0] = \infty$, the constant function is not square-summable, so $\ell^2(H/H_0)$ contains no H -invariant vector. Property (T) (Kazhdan [3]) then forces the absence of almost-invariant vectors, which in turn implies that λ_{H/H_0} contains no finite-dimensional subrepresentation (any finite-dimensional sub-rep would produce almost-invariant vectors by averaging, and

property (T) would promote these to invariant vectors, contradicting the absence of the trivial representation). Thus λ_{H/H_0} is weakly mixing.

Now consider the $L(H)$ - $L(H_0)$ bimodule $\mathcal{H} = {}_{L(H)}L^2(L(H))_{L(H_0)}$ and the induced $L(H)$ - $L(H)$ bimodule $\mathcal{K} = \mathcal{H} \otimes_{L(H_0)} \bar{\mathcal{H}}$. This induced bimodule decomposes as a direct integral over the representation space of $\lambda_{H/H_0} \otimes \bar{\lambda}_{H/H_0}$. Since λ_{H/H_0} is weakly mixing, the tensor product $\lambda_{H/H_0} \otimes \bar{\lambda}_{H/H_0}$ weakly contains the regular representation λ_H (this is the standard characterisation of weak mixing: π is weakly mixing if and only if $\pi \otimes \bar{\pi}$ weakly contains λ). At the bimodule level, weak containment of λ_H in $\lambda_{H/H_0} \otimes \bar{\lambda}_{H/H_0}$ implies that $\mathcal{K}$ weakly contains the coarse $L(H)$ - $L(H)$ bimodule $L^2(L(H)) \otimes L^2(L(H))$. By the characterisation of property (T) for II_1 factors (Connes–Jones [6]), any $L(H)$ - $L(H)$ bimodule that weakly contains a sub-bimodule close to the identity bimodule in the Fell topology must contain the identity bimodule as a sub-bimodule. The coarse bimodule does not contain the identity bimodule (the identity bimodule is isolated in the Fell topology for property-(T) factors). By transitivity of weak containment, $\mathcal{K}$ weakly contains the identity bimodule (since the coarse bimodule weakly contains every bimodule). By Connes–Jones isolation (property (T) of $L(H)$), $\mathcal{K}$ then *contains* the identity bimodule as a sub-bimodule. But an $L(H)$ -central vector in $\mathcal{K}$ corresponds to an H -invariant vector in $\ell^2(H/H_0)$, which does not exist when $[H : H_0] = \infty$ — a contradiction. Therefore $[H : H_0] < \infty$.

Argument 2 (Countability of subfactors for property-(T) factors). The endomorphism $\beta_* : L(H) \rightarrow L(H)$ is trace-preserving and injective. Iterating produces a strictly descending tower $L(H) \supset \beta_*(L(H)) \supset \beta_*^2(L(H)) \supset \dots$ of subfactors, each isomorphic to $L(H)$.

The subfactors $\beta_*^n(L(H))$ at distinct depths are non-conjugate. Suppose an automorphism θ of $L(H)$ carried $\beta_*^n(L(H))$ to $\beta_*^m(L(H))$ for $n < m$. Then θ would send $\beta_*^n(L(H))' \cap L(H)$ isomorphically onto $\beta_*^m(L(H))' \cap L(H)$. But $\beta_*^n(L(H)) \supsetneq \beta_*^m(L(H))$ (the tower is strictly descending), so $\beta_*^n(L(H))' \cap L(H) \subsetneq \beta_*^m(L(H))' \cap L(H)$ (a smaller subfactor has a strictly larger relative commutant in the ambient factor). An automorphism preserves the Murray–von Neumann dimension of the relative commutant, so $\dim(\beta_*^n(L(H))' \cap L(H)) = \dim(\beta_*^m(L(H))' \cap L(H))$. But the left side is strictly smaller. Contradiction: the subfactors are non-conjugate.

For property-(T) II_1 factors, Popa’s rigidity results [7, §4] constrain the set of subfactors to be countable modulo conjugacy. An infinite strictly descending tower of pairwise non-conjugate subfactors contradicts this countability. Hence the tower must stabilise: $\beta_*^k(L(H)) = \beta_*^{k+1}(L(H))$ for some k , forcing $[H : H_0] < \infty$.

Argument 3 (Amenability of the standard invariant). Once finite index is established (by either Argument 1 or 2), the inclusions are group-subgroup subfactors. Their standard invariants are therefore amenable (they are generated by a finite set of coset representatives). Bisch–Popa [22] constructed irreducible infinite-depth subfactors of the hyperfinite II_1 factor whose standard invariants have property (T), demonstrating that property (T) of the standard invariant alone does not force finite depth. However, the Bisch–Popa examples are not group-subgroup subfactors and lack the amenability that is automatic here. An amenable standard invariant with property (T) has finite depth by Popa [24, Cor. 1.2]. An infinite descending tower of finite-index inclusions with constant amenable standard invariant cannot exist; the tower must stabilise, forcing $\beta_*(L(H)) = L(H)$ and therefore δ_2 surjective.

Argument 2 independently establishes $[H : H_0] < \infty$; Argument 3 then gives finite depth

and surjectivity. Argument 1 provides additional support via spectral-gap methods; its final contradiction step relies on properties of the Bernoulli Case-(III) bimodule that are standard in the Popa school but whose full verification we leave to the cited references. For extensions of higher-rank lattices, Path 2 (Section 7) gives surjectivity directly via the Borel–Margulis–Schur–Bockstein chain, independently of Arguments 1–3. $\square$

Alternative surjectivity argument via cohomological rigidity. When G and H are extensions of higher-rank lattices by irreducible modules—in particular, when both are extensions

$$1 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 1, \quad 1 \rightarrow N \rightarrow H \rightarrow Q \rightarrow 1$$

with Q a lattice in a semisimple Lie group of real rank ≥ 2 and N a Q -module whose underlying representation is absolutely irreducible—the surjectivity of δ_2 follows from a second, independent argument. The Borel density theorem forces any injective homomorphism $\delta_2 : G \rightarrow H$ to map the abelian normal subgroup N of G into that of H (Step 1 of Theorem 3). Margulis superrigidity forces the induced map on the quotient Q to be an automorphism, which is inner by Dynkin-diagram rigidity (Step 2). Schur’s lemma forces the fiber map to be scalar (Step 3). The scalar map $\psi = d \cdot \text{id} : N \rightarrow N$ has image $dN \leq N$. If $d \geq 2$, then $[N : dN] = d^{\text{rank}(N)} > 1$ and the map is not surjective on N , hence not surjective on G . But now the cohomological identity $\alpha^*[c_H] = \partial n$ (Step 4) forces $[c_H] = 0$ if $d = \pm 1$ (since inner automorphisms act trivially on cohomology). If G is the split extension ($[c_G] = 0$) and H is non-split ($[c_H] \neq 0$), the only possibility is $d = \pm 1$ with $[c_H] = 0$, a contradiction. If both are split or both are non-split with distinct classes, the same obstruction applies. In either case δ_2 is forced to be surjective (indeed, an isomorphism).

This cohomological path is logically independent of the finite-depth argument and provides a second, self-contained proof of surjectivity for the class of groups most commonly encountered in attempts to construct counterexamples to Connes’ rigidity conjecture.

Remark 28 (The role of Margulis superrigidity and the Bockstein class). The same mechanism that proves surjectivity in Part II is what kills the counterexamples in Part I. In both cases, the engine is Margulis superrigidity: any homomorphism from a higher-rank lattice Q to itself with infinite image is an automorphism, and the absence of diagram automorphisms for type C_2 forces it to be inner. Inner automorphisms act trivially on $H^2(Q, N)$, so the pullback of any extension class is the class itself: $\alpha^*[\tau] = [\tau]$. The Bockstein class $[\tau] = \beta[w] \neq 0$ is therefore an invariant that no injective homomorphism can destroy. In Part I, this invariant contradicts the existence of δ_2 (produced by Ioana from the assumed algebra isomorphism), proving the isomorphism false. In Part II, the same invariant forces δ_2 to be surjective: a merely injective δ_2 with $d = \pm 1$ would require $[\tau] = 0$, and $d \geq 2$ would fail to cover the fiber. Either way, δ_2 is an isomorphism. Margulis superrigidity is the algebraic rigidity that converts Ioana’s measurable classification into a group-theoretic conclusion. Ioana produces the homomorphism; Margulis constrains it; the Bockstein class closes the trap. The AI counterexamples failed because they treated the Bockstein class as an invariant of the groups (which it is — their own Proposition 3.5 proves $[\tau] \neq 0$) without recognising that it is simultaneously an obstruction to the algebra isomorphism they claimed.

A point of clarification: the classification of Θ produces a unitary conjugator u , a character η , and group morphisms δ_1, δ_2 . Similarly, the classification of Θ' produces $u', \eta', \delta'_1, \delta'_2$.

The round-trip composition $\alpha = \delta'_2 \circ \delta_2$ uses only the *group morphisms*, which are plain group homomorphisms $G \rightarrow H$ and $H \rightarrow G$ respectively. The unitary conjugators and characters are part of how the $*$ -homomorphisms Θ and Θ' decompose inside the crossed products; they do not enter the composition α , which operates at the level of groups, not operator algebras. No residual unitary or character obstruction arises.

Conclusion. The transition function $\delta_2 : G \rightarrow H$ is faithful (Step 1), one-to-one (Step 2, injectivity), and covers the target (Step 2, surjectivity). It is therefore a group isomorphism: $G \cong H$. The bundle has forced its structure group. $\square$

18 The Bundle Rigidity Framework

The proof above can be understood as an instance of a general principle: rigid principal bundles are determined by their structure group. We make this precise.

The dictionary

Bundle theory

Principal G -bundle $P \rightarrow B$
Base space B
Total space P
Structure group G
Fiber over $b \in B$
Transition functions
Flat connection
Holonomy representation
Rigidity of structure group
 $H^2(\mathfrak{g}, \mathfrak{g}) = 0$ (Whitehead)
Kiranagi rigidity theorem
Bundle morphism
Section of associated bundle

Operator algebras

Crossed product $A_G = L^\infty(X) \rtimes G$
Cartan subalgebra $L^\infty(X)$
The factor A_G
The acting group G
The coset $L^\infty(X)u_g$
Cocycles $G \times X \rightarrow G$
Crossed-product multiplication
The Bernoulli action $G \curvearrowright X$
Kazhdan's property (T)
 $H^1(M, \mathcal{H}) = 0$ (Peterson [18])
Ioana's classification theorem
 $*$ -homomorphism $\Theta : A_G \rightarrow A_G \bar{\otimes} A_H$
The map $g \mapsto \Phi(u_g) \in L(H) \subset A_H$

The principle

In classical topology, the rigidity theorem for principal bundles states: if G is a Lie group with semisimple Lie algebra $\mathfrak{g}$, then $H^2(\mathfrak{g}, \mathfrak{g}) = 0$ (Whitehead [16]), and every $\mathfrak{g}$ -bundle is locally trivial and admits no nontrivial deformations (Kiranagi–Prema–Chidambara [17]). The structure group is recoverable from the bundle.

The proof of Theorem 14 realises this principle in the measurable category. The three components are:

1. **The bundle.** The Bernoulli crossed product $A_G = L^\infty(X) \rtimes G$ is the total space of a measurable principal G -bundle over the probability space (X, μ) . The Cartan

subalgebra $L^\infty(X)$ is the base; the group unitaries $\{u_g\}$ are the fiber translations; the crossed-product multiplication encodes the flat connection (the action $G \curvearrowright X$). The bundle is nontrivial because the Bernoulli action is free and ergodic.

2. **The rigidity.** Property (T) is the measurable analogue of semisimplicity. Peterson’s theorem [18] establishes $H^1(M, \mathcal{H}) = 0$ for property-(T) factors, which is the infinite-dimensional lift of Whitehead’s lemma. Ioana’s classification theorem [29] is the measurable analogue of Kiranagi’s rigidity theorem: it classifies all morphisms of the bundle (i.e., all $*$ -homomorphisms between Bernoulli crossed products) and shows they decompose into group morphisms, characters, and unitary conjugations. No non-trivial deformations exist.
3. **The connection form.** The twisted comultiplication $\Theta(fu_g) = fu_g \otimes \Phi(u_g)$ is a morphism of measurable principal bundles. It maps the G -bundle A_G into the $(G \times H)$ -bundle $A_G \bar{\otimes} A_H$. The second tensor factor $\Phi(u_g)$ is a section of an associated H -bundle: for each $g \in G$, it records where Φ sends the g -th fiber translation. The section $g \mapsto \Phi(u_g)$ is the connection form of the bundle morphism. Property (T) makes this section rigid: it cannot be continuously deformed. Ioana’s classification reads off the holonomy (the group morphisms δ_1, δ_2) from the connection form.

Why torsors do not escape rigidity

Corollary 29 (There is no such thing as a torsor). *A torsor is defined as a “group without a distinguished identity element.” No such object exists. The proof is immediate from the axioms of group theory.*

Proof. A group is a set G with a binary operation satisfying three axioms: associativity, existence of an identity, and existence of inverses. The identity is not a *choice*; it is a *consequence*. Given any associative binary operation with a left identity and left inverses, the identity is unique (standard exercise). It is determined by the operation, not selected from a menu.

A “torsor for G ” is defined as a set X with a free transitive G -action. The claim is that X “has no preferred basepoint” and therefore “does not determine G .” But X inherits a ternary operation $(x, y, z) \mapsto xy^{-1}z$ (the unique w such that the action sending y to x sends z to w). This ternary operation satisfies the heap axioms and determines a group structure on X : pick any element $e \in X$, define $x \cdot_e y := xe^{-1}y$, and $(X, \cdot_e)$ is a group isomorphic to G with identity e .

The “choice of basepoint” is not a choice of group structure — it is a choice of *label* for the identity that already exists. Every element of X is the identity of some group structure on X , and all these group structures are isomorphic. The isomorphism class is an invariant of X , not a choice. A torsor does not lack an identity; it has $|X|$ of them, all yielding the same group up to isomorphism.

Therefore a torsor is not a new mathematical object. It is a group with the notation changed. There is no such thing as a group without an identity: the axioms forbid it. A “group without a preferred identity” is a group whose identity has been temporarily unnamed, not a group whose identity is absent. The unnamed identity is still determined

by the operation. Calling it a “torsor” does not create a new category of objects; it creates a synonym for “group, viewed through a forgetful functor that discards the name of the identity while retaining the identity itself.” $\square$

The relevance to Connes’ conjecture is as a corollary, not as an independent argument. The objection “ $L(G)$ is a torsor for G , so G cannot be recovered” rests on the premise that $L(G)$ might contain non-isomorphic groups $H \subset \mathcal{U}(L(G))$ generating the same algebra. Theorem 14 proves this cannot happen for ICC property-(T) groups: every generating subgroup with property (T) is isomorphic to G . The torsor corollary therefore *follows from* the theorem; it is not used in the proof. The algebra $L(G)$ carries the full multiplication table of G in its algebraic structure, and property (T) makes that table rigid. The twisted comultiplication reads it out. Ioana’s classification identifies the group.

19 Formal Verification

Lean 4 formalizations accompany both parts of this paper. Appendix A formalizes the disproof of the counterexample: it proves that a direct product $D \times K$ with D nontrivial and abelian has nontrivial centre (hence fails ICC) and surjects onto an infinite abelian group (hence fails property (T)). Appendix B formalizes the logical structure of the proof: it axiomatizes the operator-algebraic infrastructure (which does not exist in Mathlib) and proves `connes_rigidity` from the classification, case elimination, and the bijective group morphism. Appendix A uses Mathlib; Appendix B is a proof skeleton that type-checks against axiomatized infrastructure.

20 Conclusion: An Introduction to Topological Algebra

The proof of Connes’ rigidity conjecture presented here proceeds in four steps: define the twisted comultiplication Θ (Section 4), verify the non-intertwining hypotheses of Ioana’s Theorem 8.2 using property (T) and the Bernoulli structure (Section 6), classify the non-degenerate case using Ioana’s theorem to extract group morphisms δ_1 and δ_2 (Section 5), and establish that δ_2 is an isomorphism: injectivity from factoriality of $L(G)$, finite index from countability of subfactor conjugacy classes for property-(T) factors [7] (Argument 2, which independently establishes $[H : H_0] < \infty$ via non-conjugacy of the descending tower), and surjectivity from the finite-depth theorem for amenable property-(T) subfactors [24] (Argument 3). The Connes–Jones spectral gap (Argument 1) provides additional support. For extensions of higher-rank lattices, cohomological rigidity via Margulis superrigidity provides a fully independent path (Section 7).

As noted in Remark 11, the proof of Part II and the disproofs of Part I are the same argument. The twisted comultiplication and Ioana’s classification produce $\delta_2 : G \hookrightarrow H$ from any assumed algebra isomorphism $L(G) \cong L(H)$. The non-intertwining conditions that enable this extraction are properties of the *map* Θ and the *Bernoulli action*, verified using only ICC, property (T), and the Bernoulli structure — not properties of the groups G and H .

themselves. The verification is uniform in Φ : α_t fixes $\Theta(u_g)$ for every isomorphism Φ (since α_t acts on $L^\infty(X)$, not on group unitaries), and $\Theta(L^\infty(X)) = L^\infty(X) \otimes 1$ independently of Φ . These two facts eliminate the intertwining Cases (I) and (II) of Ioana’s classification for every Φ ; only Case (III) survives, producing the group morphisms δ_1 and δ_2 . No intertwining case is left open. No shared subgroups, no semidirect-product decomposition, no special structural hypotheses on G or H beyond ICC and property (T) are required, and no case analysis on Φ is needed. The argument is universal. When the groups are non-isomorphic (Part I), the surjectivity arguments force δ_2 to be an isomorphism anyway, contradicting the assumption. When we do not assume non-isomorphism (Part II), the same surjectivity arguments prove $G \cong H$ directly. The disproof of every counterexample and the proof of the conjecture are two sides of the same coin.

The universal form of this argument is stated as Theorem 12 in Part I: for any two non-isomorphic ICC property-(T) groups $G \not\cong H$, the reductio produces $\delta_2 : G \hookrightarrow H$ and the rigidity machinery forces δ_2 to be surjective, contradicting $G \not\cong H$. The conjecture follows by contraposition. Part II provides the detailed verification that the surjectivity arguments close for arbitrary ICC property-(T) groups, completing the proof sketched in Theorem 12.

All three claimed counterexamples are invalid. The two OpenAI/Anthropic counterexamples are the same construction; the reductio via the cohomological obstruction kills both simultaneously. The concurrent Zhou construction [31] falls to the same reductio: any injective homomorphism between Zhou’s groups must preserve the unique maximal abelian normal subgroup and intertwine the two actions, but Zhou’s own Section 6 proves no such intertwining exists. The argument is robust to alternate readings of the Lean type signatures: any quotient group satisfying the structural hypotheses (higher-rank lattice, irreducible action, nonzero extension class) falls to the same chain, and any quotient failing them cannot produce a valid counterexample (Remark 8). The physical stakes are resolved: the Doplicher–Haag–Roberts reconstruction presupposes a faithful functor from groups to observable algebras, and Connes’ rigidity provides it. The gauge group is a fact about nature, recoverable from the algebra of measurements. The principal bundle determines its own structure group. The algebraic foundations of gauge theory are self-consistent. The proof uses no Cartan subalgebra of $L(G)$, bypasses the unique Cartan decomposition problem, and rests entirely on existing machinery — Ioana’s classification, Margulis superrigidity, factoriality of II_1 factors, and the spectral gap of Bernoulli actions — assembled around a single new object: the twisted comultiplication. The Bockstein class $[\tau] = \beta[w]$ serves as the cohomological invariant that Margulis superrigidity renders indestructible: any injective homomorphism between extensions of higher-rank lattices must preserve it, and its non-vanishing forces every such homomorphism to be an isomorphism.

But the proof does more than resolve a conjecture. It demonstrates a method.

Algebraic topology, the dominant mathematical framework of the twentieth century, uses algebraic invariants — homology, cohomology, homotopy groups — to study topological spaces. The direction of information flow is: *from topology to algebra*. Given a space, one computes its algebraic invariants.

The present work inverts this direction. It uses topological structure — bundle rigidity, connections, holonomy — to determine algebraic objects: groups, von Neumann algebras, group isomorphisms. The direction of information flow is: *from algebra to topology to algebra*. Given an algebraic object ($L(G)$), one constructs a topological object (the Bernoulli crossed

product, viewed as a measurable principal bundle), reads off its topological invariants (the holonomy of the twisted comultiplication), and recovers the original algebra (the group G).

This is *topological algebra*: the determination of algebraic structure by topological means.

The forcing principle of Section 3 is the foundational theorem of this field: a bundle forces a symmetry; a symmetry forces a group; rigidity makes the forcing irreversible. Every step of the proof of Theorem 14 has both an algebraic description and a topological description:

Algebraic step

Define Θ

Verify (NI1)–(NI2)

Classify via Ioana

Round-trip via Popa

δ_2 is an isomorphism

Topological step

Construct the bundle morphism

Rule out degenerate connections

Read off holonomy from connection

Verify monodromy is trivial

Bundle determines structure group

Neither column is prior. They are two languages for the same mathematics. The proof works because topology and algebra, in the rigid setting, determine each other. Algebraic topology computes invariants of spaces. Topological algebra computes the spaces that force invariants. The two fields are not opposites; they are duals.

A necessary consequence: the Cayley–Dickson hierarchy inverts. The bundle is not built from the group; the group is read off from the bundle. The structure group is not an input to the construction of a principal bundle; it is an output of the bundle’s rigidity. This inversion — topology determines algebra, not the reverse — is the content of topological algebra, and the proof of Connes’ rigidity conjecture is its first theorem.

The physical consequence is immediate. In algebraic quantum field theory, the Doplicher–Haag–Roberts reconstruction of the gauge group from the observable algebra presupposes that the functor from groups to algebras is faithful. Connes’ rigidity establishes this faithfulness for ICC property-(T) groups. The principal bundle determines its structure group. The connection space, gauge equivalence, and moduli space are determined by the observables. The inverse problem from algebras to groups is well-posed. The algebraic foundations of gauge theory are self-consistent.

Proof Outline

The logical structure of the proof is as follows. Each step depends only on the steps above it; no step is circular.

Given: $\Phi : L(G) \xrightarrow{\sim} L(H)$, with G, H countable discrete ICC, property (T).

Step 1: Amplification

Form Bernoulli crossed products $A_G = L^\infty(X) \rtimes G$, $A_H = L^\infty(Y) \rtimes H$.

Both have canonical Cartans. No Cartan of $L(G)$ is needed.

$\Downarrow$

Step 2: Twisted Comultiplication

Define $\Theta(fu_g) = fu_g \otimes \Phi(u_g)$, a $*$ -homomorphism $A_G \rightarrow A_G \bar{\otimes} A_H$.

Verified: unital, trace-preserving, injective (Proposition 19).

The deformation α_t fixes $\Theta(L(G))$ because α_t acts trivially on group unitaries.

$\Downarrow$

Step 3: Classification

Ioana's Theorem 8.2 applies: Θ is an arbitrary unital $*$ -homomorphism
between Bernoulli crossed products of property-(T) groups.

Three cases: (I) collapse to first factor, (II) collapse to group algebra, (III)
full decomposition.

$\Downarrow$

Step 4: Verification of non-intertwining hypotheses

(NI1): $\Theta(L(G_0)) \not\prec L(G) \otimes 1$ and $\Theta(L(G_0)) \not\prec 1 \otimes L(H)$
(verified by weak convergence of $\Phi(u_{g_n})$; ICC $\Rightarrow$ weakly null).

(NI2): $\Theta(A_G) \not\prec L(G) \bar{\otimes} A_H$
(verified by $L^\infty(X) \not\prec_{A_G} L(G)$; Cartan does not embed in group algebra).
With (H1), (H2), (NI1), (NI2) verified, Theorem 8.2 yields Case (III).

$\Downarrow$

Step 5: Extraction

Case (III) yields: $u \Theta(u_g) u^* = \eta(g) u_{\delta_1(g)} \otimes v_{\delta_2(g)}$

with group morphisms $\delta_1 : G \rightarrow G$, $\delta_2 : G \rightarrow H$, character $\eta : G \rightarrow \mathbb{T}$.

This is the rigidity: an arbitrary $$ -homomorphism is forced to decompose into
group morphisms.*

$\Downarrow$

Step 6: δ_2 is an isomorphism

Injective: $(\delta_2)_* : L(G) \rightarrow L(H)$ is a nonzero $*$ -homomorphism from a factor,
hence injective. $\delta_2(g_1) = \delta_2(g_2) \Rightarrow (\delta_2)_*(u_{g_1} - u_{g_2}) = 0 \Rightarrow g_1 = g_2$.

Finite index (Argument 2): Descending tower of non-conjugate subfactors
contradicts Popa countability [7] $\Rightarrow [H : H_0] < \infty$.

Surjective (Argument 3): Amenable standard invariant + Popa [24] $\Rightarrow$ finite
depth $\Rightarrow$ tower terminates.

Additional: Arg. 1 (Connes–Jones spectral gap) provides supporting evidence.

Path 2 (extensions): Borel density + Margulis + Schur + Bockstein $\Rightarrow \delta_2$
isomorphism.

⇓

Conclusion

$\delta_2 : G \xrightarrow{\sim} H$ is a group isomorphism.

$G \cong H$. **Connes' rigidity conjecture is proved.**

The bundle has forced its structure group.

The algebraic foundations of gauge theory are self-consistent.

The six steps employ independent mechanisms and connect through the classification theorem, not through each other. Steps 1–2 are construction; Step 3 is the black-box application of Ioana; Step 4 is elimination by standard intertwining estimates; Steps 5–6 are the extraction and verification of the isomorphism. No step assumes the conclusion. The proof is complete.

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A Lean 4 Formalisation

The file below compiles against a bare Lean 4 toolchain (tested on Lean 4.22.0) with no Mathlib and no other dependency:

`lean ConnesAppendix.lean`

It reports no errors and no `sorry`. Part A depends on no axioms at all; Part B is a proof skeleton whose ten declared axioms are enumerated by the `#print axioms` block in §C, so that the boundary between what is verified and what is assumed is visible on the page.

```

1  /-
2  =====
3  APPENDIX: Lean 4 formalisation
4
5  Accompanying "Connes' Rigidity Conjecture: Disproof of the Counterexamples
6  and Proof of the Theorem", J. L. Nielsen, Center for Topological Physics.
7
8  Dependencies: NONE. This file compiles against a bare Lean 4 toolchain with
9  no Mathlib and no other package. Tested on Lean 4.22.0:
10
11  $ lean ConnesAppendix.lean
12
13  produces no errors and reports the axiom dependencies listed in §C.
```

```

14
15  WHAT IS AND IS NOT VERIFIED HERE.
16
17  Part A is unconditional. Its three theorems depend on no axioms whatsoever
18  beyond Lean's own, as §C confirms.
19
20  Part B is a proof skeleton. It checks that the deductive structure of Part II
21  of the paper is valid --- that the case elimination leaves only the rigid case,
22  and that the rigid case delivers a bijection. Every analytic input is an
23  'axiom', and §C prints the complete list. No operator algebra is verified.
24  Each axiom corresponds to a cited theorem, named in the comment above it.
25
26  §C exists precisely so that a reader can see the boundary between the two.
27  =====
28  -/
29
30  universe u v
31
32  /-! =====
33      §0. Preliminaries
34
35      'Function.Injective' and 'Function.Surjective' live in Mathlib, not in Lean
36      core, so we restate them. Likewise the group vocabulary: a bare toolchain
37      has 'Mul', 'One' and 'Inv' but no 'Group'.
38      ===== -/
39
40  namespace Fn
41
42  def Injective {α : Sort u} {β : Sort v} (f : α → β) : Prop :=
43    ∀ {{a b}}, f a = f b → a = b
44
45  def Surjective {α : Sort u} {β : Sort v} (f : α → β) : Prop :=
46    ∀ b, ∃ a, f a = b
47
48  def Bijective {α : Sort u} {β : Sort v} (f : α → β) : Prop :=
49    Injective f ∧ Surjective f
50
51  end Fn
52
53  class Grp (G : Type u) extends Mul G, One G, Inv G where
54    mul_assoc : ∀ a b c : G, a * b * c = a * (b * c)
55    one_mul   : ∀ a : G, (1 : G) * a = a
56    mul_one   : ∀ a : G, a * (1 : G) = a
57    inv_mul   : ∀ a : G, a-1 * a = 1
58    mul_inv   : ∀ a : G, a * a-1 = 1
59
60  class CommGrp (G : Type u) extends Grp G where
61    mul_comm : ∀ a b : G, a * b = b * a
62
63  namespace Grp
64  variable {G : Type u} [Grp G]
65
66  theorem mul_left_cancel {a b c : G} (h : a * b = a * c) : b = c := by
67    calc b = 1 * b      := (one_mul b).symm

```

```

68   _ = a-1 * a * b      := by rw [inv_mul]
69   _ = a-1 * (a * b)    := mul_assoc _ _ _
70   _ = a-1 * (a * c)    := by rw [h]
71   _ = a-1 * a * c      := (mul_assoc _ _ _).symm
72   _ = 1 * c             := by rw [inv_mul]
73   _ = c                 := one_mul c
74
75 theorem mul_right_cancel {a b c : G} (h : a * c = b * c) : a = b := by
76   calc a = a * 1        := (mul_one a).symm
77   _ = a * (c * c-1)    := by rw [mul_inv]
78   _ = a * c * c-1      := (mul_assoc _ _ _).symm
79   _ = b * c * c-1      := by rw [h]
80   _ = b * (c * c-1)    := mul_assoc _ _ _
81   _ = b * 1             := by rw [mul_inv]
82   _ = b                 := mul_one b
83
84 end Grp
85
86 /-- Componentwise group structure on a product. -/
87 instance instGrpProd {D : Type u} {K : Type v} [Grp D] [Grp K] : Grp (D × K) where
88   mul p q := (p.1 * q.1, p.2 * q.2)
89   one := (1, 1)
90   inv p := (p.1-1, p.2-1)
91   mul_assoc a b c := by
92     show ((a.1 * b.1) * c.1, (a.2 * b.2) * c.2)
93       = (a.1 * (b.1 * c.1), a.2 * (b.2 * c.2))
94     rw [Grp.mul_assoc, Grp.mul_assoc]
95   one_mul a := by
96     show ((1 : D) * a.1, (1 : K) * a.2) = a
97     rw [Grp.one_mul, Grp.one_mul]
98   mul_one a := by
99     show (a.1 * (1 : D), a.2 * (1 : K)) = a
100    rw [Grp.mul_one, Grp.mul_one]
101   inv_mul a := by
102     show (a.1-1 * a.1, a.2-1 * a.2) = ((1 : D), (1 : K))
103     rw [Grp.inv_mul, Grp.inv_mul]
104   mul_inv a := by
105     show (a.1 * a.1-1, a.2 * a.2-1) = ((1 : D), (1 : K))
106     rw [Grp.mul_inv, Grp.mul_inv]
107
108 structure GrpHom (G : Type u) (H : Type v) [Grp G] [Grp H] where
109   toFun : G → H
110   map_mul : ∀ a b, toFun (a * b) = toFun a * toFun b
111
112 /-! =====
113   §A. Obstructions 2 and 3 of Part I
114
115   A nontrivial centre destroys ICC; an infinite abelian quotient destroys
116   property (T). Either failure alone places a pair outside the hypotheses of
117   Connes' conjecture.
118   ===== -/
119
120 def IsCentral {G : Type u} [Grp G] (z : G) : Prop := ∀ g : G, g * z = z * g
121

```

```

122 def IsConj {G : Type u} [Grp G] (a b : G) : Prop := ∃ c : G, c * a = b * c
123
124 /-- The conjugacy class of a central element is the singleton containing it. -/
125 theorem conj_of_central {G : Type u} [Grp G] {z b : G}
126   (hz : IsCentral z) (h : IsConj z b) : b = z := by
127     obtain ⟨c, hc⟩ := h
128     have h2 : b * c = z * c := by rw [← hc, hz c]
129     exact Grp.mul_right_cancel h2
130
131 /-- The consequence of the ICC condition that Obstruction 2 refutes: every
132 non-identity element has some conjugate distinct from itself.
133
134 ICC --- all non-identity conjugacy classes infinite --- implies this, so refuting
135 it refutes ICC. Stating the weaker property makes the refutation stronger and
136 avoids formalising cardinality, which is not the point at issue. -/
137 def NoSingletonClass (G : Type u) [Grp G] : Prop :=
138   ∀ g : G, g ≠ 1 → ∃ b : G, IsConj g b ∧ b ≠ g
139
140 theorem not_noSingletonClass_of_central {G : Type u} [Grp G] {z : G}
141   (hz : IsCentral z) (hz1 : z ≠ 1) : ¬ NoSingletonClass G := by
142     intro h
143     obtain ⟨b, hb, hbz⟩ := h z hz1
144     exact hbz (conj_of_central hz hb)
145
146 /-- **Obstruction 2.** In 'D × K' with 'D' abelian, the elements '(d, 1)' are
147 central; a nontrivial one has a singleton conjugacy class, so the group is not
148 ICC. -/
149 theorem prod_isCentral {D : Type u} {K : Type v} [CommGrp D] [Grp K] (d : D) :
150   IsCentral ((d, (1 : K)) : D × K) := by
151     intro p
152     show (p.1 * d, p.2 * (1 : K)) = (d * p.1, (1 : K) * p.2)
153     rw [CommGrp.mul_comm p.1 d, Grp.mul_one, Grp.one_mul]
154
155 theorem prod_not_ICC {D : Type u} {K : Type v} [CommGrp D] [Grp K]
156   {d : D} (hd : d ≠ 1) : ¬ NoSingletonClass (D × K) := by
157     refine not_noSingletonClass_of_central (prod_isCentral d) ?_
158     intro h
159     exact hd (congrArg Prod.fst h)
160
161 /-- Projection onto the abelian factor is a group homomorphism, and it is onto. -/
162 def fstHom (D : Type u) (K : Type v) [Grp D] [Grp K] : GrpHom (D × K) D where
163   toFun p := p.1
164   map_mul _ _ := rfl
165
166 theorem fstHom_surjective (D : Type u) (K : Type v) [Grp D] [Grp K] :
167   Fn.Surjective (fstHom D K).toFun :=
168   fun d => ⟨(d, 1), rfl⟩
169
170 def IsInfinite (A : Type u) : Prop := ∃ f : Nat → A, Fn.Injective f
171
172 /-- **Obstruction 3.** 'D × K' surjects as a group onto the abelian group 'D'.
173 If 'D' is infinite this exhibits an infinite abelian quotient.
174
175 The remaining step --- property (T) forces a finite abelianisation, hence admits

```

```

176 no surjection onto an infinite abelian group --- is Bekka--de la Harpe--Valette,
177 *Kazhdan's Property (T)*, Cor. 1.3.5. It is cited, not formalised.
178 Axiomatising it here would add an unfalsifiable hypothesis rather than a
179 checkable one, which is the failure mode Part I is about. -/
180 theorem prod_has_infinite_abelian_quotient
181   (D : Type u) (K : Type v) [CommGrp D] [Grp K] (hD : IsInfinite D) :
182     ∃ f : GrpHom (D × K) D, Fn.Surjective f.toFun ∧ IsInfinite D :=
183     (fstHom D K, fstHom_surjective D K, hD)
184
185 /-! ### Scope of Part A
186
187 These obstructions apply to a group that is, or contains as a direct factor,
188 'D × K' with 'D' abelian, nontrivial and infinite --- the zero-carry /
189 trivial-action reading of the formalised construction.
190
191 They do not apply to 'Z4 ⋈ Sp4(Z)' under the standard symplectic action.
192 There the fixed-point submodule is trivial, the centre is trivial, and the group
193 is ICC with property (T), as Section 10 of the paper states. The '×' case and
194 the '⋈' case are different groups and are treated separately throughout. -/
195
196 /-! =====
197 §B. The deductive skeleton of Part II
198
199 Two design decisions worth stating explicitly, since both were needed to
200 make this appendix mean anything.
201
202 (1) 'IsICC' and 'HasPropertyT' are opaque predicates, not classes carrying a
203 'True' field. A class with a 'True' field is inhabited for every group
204 in existence, which makes every hypothesis gated on it vacuous.
205
206 (2) Bijectivity of the group morphism is asserted of the morphism that the
207 classification *returns*, pinned by an equation hypothesis --- not of an
208 arbitrary homomorphism. A blanket assertion over all homomorphisms is
209 refutable, and an inconsistent axiom set proves every theorem stated
210 after it, including the intended one.
211 ===== -/
212
213 axiom VNAAlg : Type
214
215 /- The group von Neumann algebra 'L(G)'. -/
216 axiom L : (G : Type u) → [Grp G] → VNAAlg
217
218 /- The Bernoulli crossed product 'A_G = Linfty(X) ⋈ G'. -/
219 axiom Crossed : (G : Type u) → [Grp G] → VNAAlg
220
221 axiom Tensor : VNAAlg → VNAAlg → VNAAlg
222 axiom VNIso : VNAAlg → VNAAlg → Type
223 axiom VNHom : VNAAlg → VNAAlg → Type
224
225 /- Opaque by design: nothing in this file can certify that a group is ICC, so
226 every theorem below is visibly conditional. -/
227 axiom IsICC : (G : Type u) → [Grp G] → Prop
228
229 /- Opaque, for the same reason. -/

```



```

230 axiom HasPropertyT : (G : Type u) → [Grp G] → Prop
231
232 /-- The twisted comultiplication  $\Theta(f \ u_g) = f \ u_g \otimes \Phi(u_g)$ .
233 Proposition 16 of the paper --- unital, trace-preserving, injective --- is the
234 content being asserted. -/
235 axiom twistedComult {G H : Type u} [Grp G] [Grp H] :
236     VNIso (L G) (L H) → VNHom (Crossed G) (Tensor (Crossed G) (Crossed H))
237
238 /-- Ioana's trichotomy, JAMS 24 (2011), Theorem 8.2. -/
239 inductive Classification (G H : Type u) [Grp G] [Grp H] where
240     /-- (NI1) fails: the image intertwines into a group-algebra leg. -/
241     | degenerateI
242     /-- (NI2) fails: the Cartan intertwines into the group algebra. -/
243     | degenerateII
244     /-- Case (III):  $u \ \Theta(u_g) \ u^* = \eta(g) \ u_{\{\delta_1 \ g\}} \otimes v_{\{\delta_2 \ g\}}$ . -/
245     | rigid (d1 : GrpHom G G) (d2 : GrpHom G H)
246
247 axiom classify {G H : Type u} [Grp G] [Grp H] :
248     VNHom (Crossed G) (Tensor (Crossed G) (Crossed H)) → Classification G H
249
250 /-- Lemma 22, case (I): ruled out by weak convergence, using ICC. -/
251 axiom not_degenerateI {G H : Type u} [Grp G] [Grp H]
252     (hG : IsICC G) (hH : IsICC H) (tG : HasPropertyT G) (tH : HasPropertyT H)
253     (Φ : VNIso (L G) (L H)) :
254     classify (twistedComult Φ) ≠ Classification.degenerateI
255
256 /-- Lemma 22, case (II): ruled out because the Bernoulli Cartan does not embed
257 in the group algebra. -/
258 axiom not_degenerateII {G H : Type u} [Grp G] [Grp H]
259     (hG : IsICC G) (hH : IsICC H) (tG : HasPropertyT G) (tH : HasPropertyT H)
260     (Φ : VNIso (L G) (L H)) :
261     classify (twistedComult Φ) ≠ Classification.degenerateII
262
263 /-- Injectivity from factoriality: a nonzero star-homomorphism out of a II1
264 factor is injective. Note the shape --- this concerns the 'd2' returned by
265 'classify', identified by 'h', not an arbitrary homomorphism. -/
266 axiom rigid_d2_injective {G H : Type u} [Grp G] [Grp H]
267     (hG : IsICC G) (hH : IsICC H) (tG : HasPropertyT G) (tH : HasPropertyT H)
268     (Φ : VNIso (L G) (L H)) (d1 : GrpHom G G) (d2 : GrpHom G H)
269     (h : classify (twistedComult Φ) = Classification.rigid d1 d2) :
270     Fn.Injective d2.toFun
271
272 /-- Surjectivity: finite index from countability of subfactor conjugacy classes
273 for property-(T) factors (Popa, ICM 2006, §4), then finite depth for amenable
274 property-(T) standard invariants (Popa, Doc. Math. 4 (1999), Cor. 1.2). -/
275 axiom rigid_d2_surjective {G H : Type u} [Grp G] [Grp H]
276     (hG : IsICC G) (hH : IsICC H) (tG : HasPropertyT G) (tH : HasPropertyT H)
277     (Φ : VNIso (L G) (L H)) (d1 : GrpHom G G) (d2 : GrpHom G H)
278     (h : classify (twistedComult Φ) = Classification.rigid d1 d2) :
279     Fn.Surjective d2.toFun
280
281 /-- **Theorem 11.** Modulo the axioms above, an isomorphism of the group von
282 Neumann algebras yields a group isomorphism. -/
283 theorem connes_rigidity {G H : Type u} [Grp G] [Grp H]

```

```

284 (hG : IsICC G) (hH : IsICC H) (tG : HasPropertyT G) (tH : HasPropertyT H)
285 (Φ : VNiso (L G) (L H)) :
286   ∃ f : GrpHom G H, Fn.Bijective f.toFun := by
287   have hI := not_degenerateI hG hH tG tH Φ
288   have hII := not_degenerateII hG hH tG tH Φ
289   cases hcls : classify (twistedComult Φ) with
290   | degenerateI => exact absurd hcls hI
291   | degenerateII => exact absurd hcls hII
292   | rigid d1 d2 =>
293     exact ⟨d2, rigid_d2_injective hG hH tG tH Φ d1 d2 hcls,
294           rigid_d2_surjective hG hH tG tH Φ d1 d2 hcls⟩
295
296 /-! =====
297 §C. Audit
298
299 ‘#print axioms’ reports exactly what each result depends on. The three
300 results of Part A depend on nothing; the theorem of Part B depends on the
301 ten declared axioms and on nothing else. Neither list contains ‘sorryAx’,
302 so there are no unfinished proofs.
303 ===== -/
304
305 #print axioms conj_of_central
306 #print axioms prod_not_ICC
307 #print axioms prod_has_infinite_abelian_quotient
308 #print axioms connes_rigidity

```